applied sciences

MDPI

Article

Research on Earthquake Demolition Rescue Robot Design Based on UXM–Kano–QFD Framework

Wei Peng, Yuqi Xia, Yue Han, Haiqiang Wang *, Yang Tang, Xinyu Liu and Yexin Chen

School of Industrial Design, Hubei University of Technology, Wuhan 430068, China; xiayuqi1216@outlook.com (Y.X.); tang99931@outlook.com (Y.T.)
* Correspondence: 20131026@hbut.edu.cn

Abstract

This study presents an integrated design methodology for earthquake demolition rescue robots by combining UXMs, Kano, and QFD to improve design rationality and performance in extreme rescue scenarios. It addresses key gaps in existing approaches, particularly the lack of systematic experiential data acquisition, quantitative requirement analysis, and effective design translation. UXMs are applied to reconstruct critical task scenarios and identify high-load nodes and user experience variations. The Kano model is used to prioritise and classify user requirements, which are then translated into engineering characteristics through QFD. Based on this framework, a conceptual robot design is developed using the FBS model and evaluated through process-level simulation and usability assessment. The results demonstrate that the proposed method enables structured requirement transformation and supports traceable design decisions. Simulation indicates the consistency of task workflows and coordination among functional modules at the process level. A System Usability Scale score of 80.22 indicates a relatively high level of perceived usability at the conceptual evaluation stage. The proposed methodology provides a structured and traceable conceptual design framework for earthquake rescue robots. While the current validation is based on conceptual-level evaluation, the methodology offers a traceable design pathway that may be extended to other high-risk emergency equipment with further empirical testing.

Keywords: earthquake rescue robot; user experience; conceptual design; industrial design; Kano-QFD

1. Introduction

China lies at the intersection of the Pacific Ring of Fire and the Eurasian seismic belt, ranking among the world's most seismically active nations [1]. In 2023, mainland China experienced a total of 11 earthquakes measuring 5.0 or higher on the Richter scale, including the Jishishan earthquake in Gansu, which was felt over a wide area and caused significant casualties and property damage [2]. Within the critical 72 h window following such disasters, safe and efficient demolition operations directly determine the success of life-saving efforts. With advances in robotics, rescue robots have emerged as vital equipment for earthquake relief operations due to their compact size, flexible deployment, and ability to access confined spaces, and have found widespread application in international disaster zones [3].

Significant progress has been made in the design and deployment of rescue robots through domestic and international research. International institutions such as NIST,

Check for updates

Received: 24 March 2026
Revised: 26 April 2026
Accepted: 29 April 2026
Published: 1 May 2026

Copyright: © 2026 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

Carnegie Mellon University, and Japan's NEDO emphasise enhancing operational capabilities in disaster scenarios through multimodal environmental perception, autonomous navigation, and intelligent decision-making. Multi-legged robots, flexible crawling platforms, and mobile arm systems are used to navigate rubble, conduct reconnaissance, and perform support tasks [4]. Concurrently, European and American research predominantly adopted a task chain perspective, deriving functional configurations through scenario-driven approaches to improve equipment-task alignment [5]. Domestic research has primarily focused on structural optimisation and functional expansion. For instance, Wang Ziren [6] proposed an optimised design for demolition equipment based on biomimetic structures; Zhao Zhiqiang [7] constructed a framework for all-terrain rescue robots from the perspective of demand and functional evolution; and Li Xiaojie et al. [8] employed AHP, QFD, and TRIZ to resolve multiple engineering conflicts within rescue equipment. While these studies have advanced the overall development of earthquake rescue equipment, systemic shortcomings persist in extreme demolition scenarios.

Methodologically, international research increasingly recognises the importance of the 'experience-demand-design' chain in complex equipment development [9]. User experience maps (UXMs) are commonly employed in healthcare, safety management, and service systems to chart multi-stage task pathways and experience fluctuations [10]. However, UXMs remain underutilised in disaster rescue—particularly demolition operations—limiting their capacity to quantify experiential data under extreme conditions. While the Kano model is widely employed for satisfaction classification and requirement modelling, its application to weighting requirements for emergency equipment is still in its exploratory phase [10]. Quality Function Deployment (QFD) serves as a systematic translation tool for converting requirements into engineering characteristics within Western manufacturing and high-reliability equipment design. However, its use in China's rescue equipment sector remains fragmented, lacking integration with front-end experience analysis tools. Scenario construction and experience reproduction methodologies, though employed in disaster medicine and military simulation, have seen limited application in the process-oriented design of rescue equipment. Overall, while domestic and international methodological research possesses distinct strengths, there remains a lack of an integrated methodological chain capable of spanning the entire process—from experiential insights and demand quantification to engineering translation. This deficiency is particularly pronounced for the complex, extreme scenario of breaching operations, where no systematic, verifiable design framework has yet been established.

Despite long-term progress in structural design, algorithms, and modularisation, systematic research addressing demolition—a high-risk, critical phase—remains inadequate [11,12]. Existing studies have typically focused on structural solutions or individual functional modules, lacking a comprehensive analysis of the experiential characteristics, requirement structures, and workload properties of the entire demolition task [13]. Concurrently, the non-replicable nature and extreme conditions of disaster sites make it difficult to obtain authentic experiential data through traditional interview methods [14]. Rescuers' behaviours, risk perceptions, and operational loads under high-noise, low-visibility, dusty, and pressurised conditions prove difficult to accurately replicate. This leads to 'function-demand misalignment' in design, limiting equipment usability and decision reliability [15].

Addressing these research gaps, this study focuses on the task chain during the demolition phase of urban earthquake rescue operations. It proposes an integrated design methodology framework combining UXMs, Kano, QFD, and FBS to achieve a systematic closed-loop process from experiential insights and requirement quantification to design translation and output. The objective is to construct a systematic design methodology for complex demolition scenarios, validating design decision logic and methodological

processes [16]. This provides a replicable design workflow for experiential data acquisition, requirement modelling, and engineering translation for earthquake rescue equipment, offering a reference for enhancing equipment usability, task adaptability, and emergency response capabilities [17].

The principal contributions of this study are as follows:

(1) A demolition scenario experience collection method is developed based on scenario construction and experience re-enactment. This method addresses the loss of experiential information caused by the irreproducibility of post-disaster environments and enhances the authenticity of extreme scenario experience data.
(2) An HML′ optimised requirement aggregation model is constructed. Through structured extraction of pain points, high-frequency operations, and experience fluctuation zones within the UXM, hierarchical identification of demolition task requirements is achieved.
(3) A quantitative translation mechanism linking experience, requirements, and design specifications is developed. The Kano requirement importance framework is combined with the QFD quality house matrix to form an engineering operational requirement weighting system, thereby enhancing transparency and objectivity in mapping requirements to design specifications.
(4) A conceptual design for a demolition rescue robot is proposed and evaluated at the conceptual level. The structural design is derived using the FBS (function–behaviour–structure) model, and process-level evaluation is conducted through Stateflow simulation and SUS, providing preliminary evidence for the traceability and conceptual applicability of the proposed methodology chain.

The remainder of this paper is organised as follows. Section 2 reviews related work on rescue robot design methodologies and identifies the research gap. Section 3 presents the integrated UXM–Kano–QFD methodology framework. Section 4 describes the UXM-based experience insight analysis, including participant recruitment, scenario design, and requirement extraction. Section 5 details the Kano-based user requirements classification and satisfaction analysis. Section 6 presents the QFD-based design requirement mapping and weighting calculation. Section 7 derives the conceptual design using the FBS model. Section 8 reports the process-level simulation and usability evaluation results. Section 9 discusses the findings, methodological implications, and limitations. Finally, Section 10 presents the conclusions and directions for future research.

2. Literature Review

2.1. Rescue Robot Design and Methodology

Research on earthquake rescue robots has progressed substantially over the past two decades. Early studies have concentrated on mechanical design and locomotion capabilities, with tracked and multi-legged platforms developed for rubble traversal and confined-space access. More recent work has shifted towards intelligent perception, autonomous navigation, and modular tool systems that can be reconfigured for diverse disaster scenarios. Internationally, institutions such as NIST, Carnegie Mellon University, and Japan's NEDO have emphasised multimodal environmental sensing and task-chain-driven functional configuration. In China, researchers have explored biomimetic structural optimisation, demand-driven functional evolution, and multi-method integration using AHP, QFD, and TRIZ to resolve engineering conflicts. Despite these advances, systematic design approaches that explicitly incorporate user experience data from extreme demolition scenarios remain scarce.

2.2. User Experience Mapping and Requirement Elicitation

User experience maps (UXMs) have been widely adopted in healthcare, service design, and safety management to visualise multi-stage task pathways, emotional fluctuations, and pain points. However, their application in disaster rescue contexts—particularly for high-risk demolition operations—remains limited. The Kano model, originally developed for consumer product satisfaction analysis, has been increasingly applied to classify and prioritise requirements in engineering contexts. Quality Function Deployment (QFD) provides a systematic matrix-based approach for translating user requirements into engineering characteristics and has been employed in manufacturing, automotive, and defence industries. Recent studies have begun to integrate Kano and QFD for product design, yet few have combined these methods with front-end experience analysis tools such as UXMs to form a complete methodology chain.

2.3. Research Gap and Positioning

Table 1 summarises representative studies in rescue robot design and related methodological frameworks. As shown, existing approaches typically address individual stages of the design process—either requirement elicitation, requirement classification, or engineering translation—without establishing a traceable chain from experiential insight to validated design output. Furthermore, none of the reviewed studies have applied UXMs to earthquake demolition scenarios or integrated it with Kano–QFD for rescue equipment design. This study addresses these gaps by proposing an integrated UXM–Kano–QFD–FBS framework that spans the entire process from experience-based requirement discovery to conceptual design validation.

Table 1. Comparative summary of representative studies on rescue robot design methodologies.

Study	Domain	UXM	Kano	QFD	FBS	Validation
Delmerico et al. [4]	SAR robotics	No	No	No	No	Conceptual
Li et al. [8]	Rescue equipment	No	No	Yes	No	AHP/TRIZ
Li and Kim [9]	Cultural heritage app design	No	Yes	Yes	No	Kano–QFD analysis
Kim [10]	Product–service systems	Yes	No	No	No	Iterative evaluation
Li and Zhao [11]	Rescue robots	No	No	No	No	FAHP/Radar
Niu et al. [17]	Underwater rescue robots	No	Yes	No	No	Digital twin
This study	Earthquake demolition	Yes	Yes	Yes	Yes	Stateflow + SUS

3. Integrated UXM–Kano–QFD Methodology Framework

The integrated UXM–Kano–QFD methodology proposed in this study comprises three progressive modules: the user experience map (UXM), the Kano demand classification model, and Quality Function Deployment (QFD). This establishes a traceable closed-loop process from user experience insights to engineering design parameters, as illustrated in Figure 1 [18].

Firstly, a UXM is a service design tool that systematically charts users' behavioural, emotional, and pain point shifts over time within specific contexts [19]. It is frequently employed for experience insights and requirement discovery in complex task scenarios, reconstructing critical situations for task decomposition. By analysing operational workflows, risk exposures, and experience fluctuations, it identifies high-load nodes and key operational touchpoints, thereby forming an initial requirement set. This module provides contextual support for requirement refinement, ensuring requirements are both contextualised and traceable.

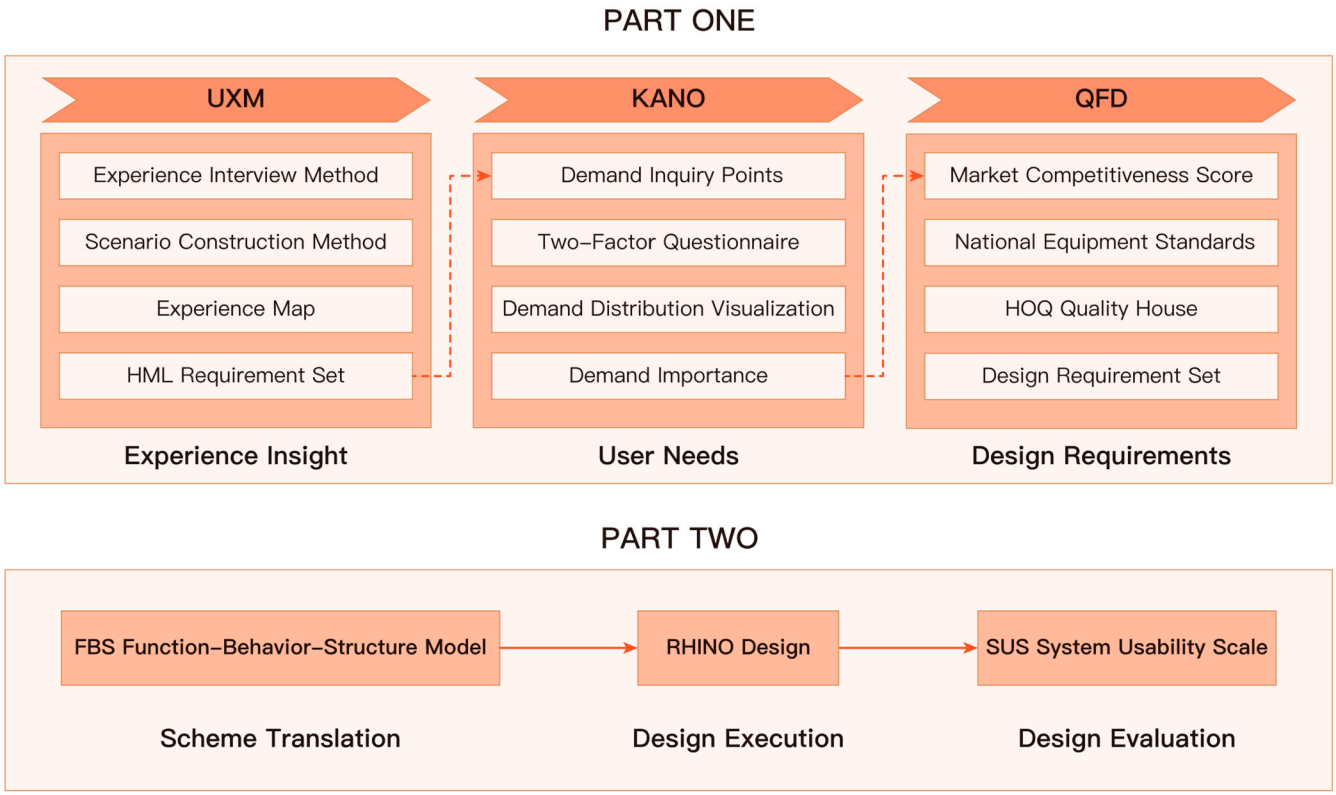


Figure 1. Research framework based on the UXM–Kano–QFD model.

Subsequently, the Kano model classifies requirements based on the non-linear characteristics of customer satisfaction, distinguishing the differential impact of different requirement attributes on satisfaction [20]. The model categorises attributes and models satisfaction based on requirements extracted from UXMs, classifying requirements into must-be, one-dimensional, attractive, and indifferent types, and quantifying their contribution through CS series metrics [21]. This step converts qualitative experiences into comparable demand weights, providing structured input for design decisions.

Finally, QFD is a product planning and decision-making method that transforms user demand systems into engineering characteristics through matrix-based approaches, establishing mappings between demands and engineering attributes [22]. Using the quality house matrix, the weightings of different demands are further translated into corresponding design requirements and technical priorities, enabling the systematic translation of demands into engineering parameters.

4. Experience Insight Analysis Based on UXMs

The user experience map (UXM) charts behavioural pathways, touchpoints, and experience fluctuations within complex tasks, enabling the identification of critical pain points and high-load nodes. Earthquake demolition tasks comprise sequential stages, including detection, demolition, and debris clearance, with each phase exhibiting distinct differences in operational load and risk exposure [23]. This study employs task flow diagrams and expert interviews to decompose demolition tasks structurally, clarifying sub-task objectives and behavioural elements. This provides clear task boundaries and a data foundation for constructing UXM scenarios and subsequent requirement extraction.

4.1. Subject Recruitment and Scenario Experience Design

To obtain authentic, traceable data on extreme demolition experience, this study recruited 12 male participants with earthquake demolition rescue experience through

purposive sampling via online interviews and field visits. Participants were aged between 25 and 45 years (mean = 34.2, SD = 5.8), each with a minimum of 3 years of professional experience in earthquake rescue demolition operations. The inclusion criteria required: (a) active service in earthquake rescue teams or related emergency response units; (b) direct operational experience in structural demolition during post-earthquake rescue; and (c) physical fitness certification for high-risk rescue operations. The gender composition reflects the demographic reality of earthquake demolition rescue teams in China, which are predominantly staffed by male personnel. This study was approved by the Research Ethics and Technology Security Committee of the School of Industrial Design, Hubei University of Technology (approval date: 1 January 2025; validity period: 1 January 2025 to 31 December 2026). Written informed consent was obtained from all participants prior to data collection. All personal data were anonymised and stored in accordance with institutional data protection protocols. Within a quasi-realistic earthquake demolition environment constructed by the research team, participants completed the full operational sequence—from location identification and demolition to spatial rescue—following a predefined task chain. The experiential process encompassed task briefing and environmental verification, task execution within simulated scenarios, immediate post-task verbal reflection (combining think-aloud protocols and retrospective interviews), annotation of critical experiential nodes, and subjective workload assessments [24]. This methodology enabled the simultaneous collection of behavioural, perceptual, and experiential data, establishing a reliable and structured foundation for subsequent UXM development [25].

4.2. Construction of Extreme Demolition Scenarios and Multidimensional Experience Data Collection

Given the irreproducibility of real earthquake demolition scenarios, this study employs a 'scenario construction + experiential recreation' approach to simulate extreme demolition environments. The environment includes typical characteristics such as collapsed debris, confined spaces, low-light conditions, dust obstructions, and high noise levels. Perceptual interference is controllably replicated through audiovisual systems and structural simulation apparatus. During the experiential execution, four categories of core data were collected as shown in Table 2. All data were synchronously recorded based on time series and can be directly mapped to task nodes within the UXM [26].

Table 2. Four categories of core data collected during the experiential execution process.

Data Type	Indicator Content	Collection Method
Behavioural Data	Operation sequence, action path, Operational rhythm, tool switching frequency	Video recording, motion observation
Load Data	Task difficulty, physical exertion, cognitive load (NASA-TLX)	Scales and real-time assessment
Emotional/Risk Perception	Tension, risk judgement, stress fluctuations	Verbal recall, expression and behavioural recording
Environmental Disturbance Factors	Lighting variations, noise intensity, dust obstruction	Environmental monitoring and manual logging

4.3. Construction and Analysis of the Earthquake Demolition Rescue User Experience Map (UXM)

Based on synchronous collection of multidimensional experiential data, this study constructed an earthquake demolition rescue user experience map, as illustrated in Figure 2. This map presents participants' behavioural pathways, touchpoint characteristics, and experience fluctuations throughout the complete task chain. The UXM sequentially unfolds the three consecutive phases—localisation, demolition, and spatial rescue—along the horizontal axis representing task stages. The vertical axis integrates multi-source data,

including behavioural performance, operational load, risk perception, and emotional fluctuations, to depict the dynamic evolution of experience throughout the task.

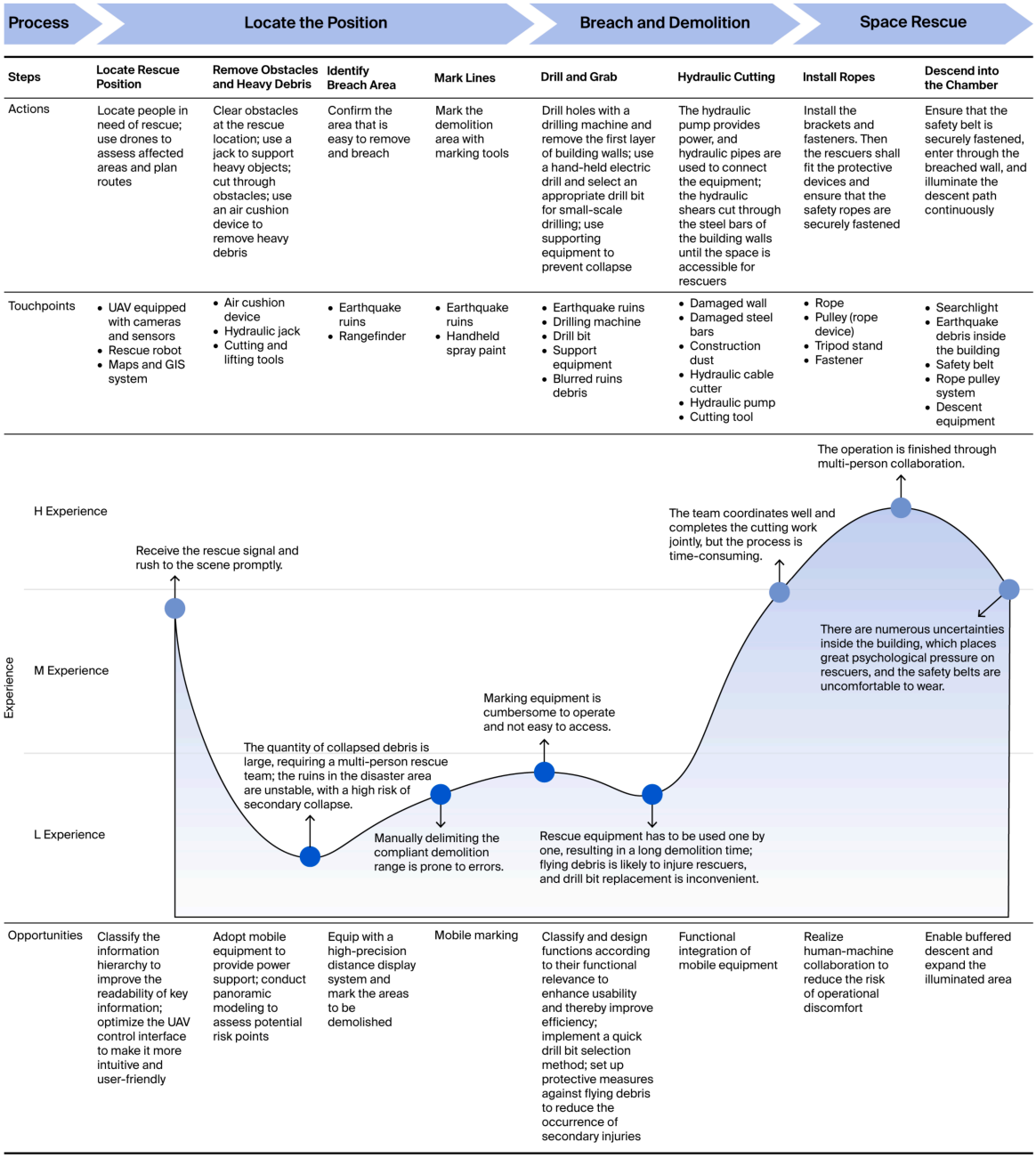


Figure 2. User experience map for earthquake demolition rescue operations.

UXM analysis reveals distinct phased variations in the experiential structure of demolition rescue tasks. During the positioning phase, tasks are frequently constrained by inadequate ranging capabilities and insufficient spatial information, resulting in inefficient target confirmation and disrupted operational rhythm [27]. In the demolition phase, limited equipment autonomy and frequent tool switching lead to sustained increases in operational load, while dust obstructions and debris spray significantly heighten secondary injury risks. During spatial rescue, reduced visibility further increases cognitive pressure and emotional fluctuations among participants. These factors collectively form the Low Experience Zone (L-zone) within the UXM. Its concentrated location and pronounced characteristics identify it as a critical pain point and high-load node within the demolition task.

In summary, the experience inflexion points and high-frequency touchpoints revealed by UXMs provide direct evidence for subsequent requirement extraction. They clarify key task segments requiring priority resolution and lay a solid foundation for constructing the HML′ requirement set and conducting Kano–QFD quantitative analysis.

4.4. Extraction and Construction of the HML′ Requirement Set Based on UXMs

Building upon the contact point structure analysis within the UXM framework, this paper identifies a primary set of requirements across three dimensions: high-load nodes, high-frequency operation contact points, and experience fluctuation intervals [28]. To systematically transform user experience information into engineering requirements, a hierarchical classification method (H/M/L) is proposed based on the user experience map (UXM) to structurally process these requirements. Specifically, M (medium) denotes requirements associated with high-frequency operations or critical supporting touchpoints; H (high) signifies key requirements arising from high-load or high-risk touchpoints; while L (low) refers to secondary requirements within poor experience zones possessing improvement potential. The criteria are as follows: high-frequency operation touchpoints (M) are operational nodes triggered ≥3 times within a single task on the UXM timeline and observed in the majority of participants. High-load nodes (H) denote experience nodes where the NASA-TLX overall load score significantly exceeds the sample mean. The poor experience zone (L) encompasses scenario segments where negative emotions or risk perceptions persist across adjacent phases, accompanied by operational impediments.

Given that L intervals typically harbour numerous latent pain points and potential improvement opportunities, this study further semantically consolidated and prioritised L interval requirements to form an optimised requirement set, L′ [29]. Ultimately, the primary requirement set H∪M was merged with the refined L′ to construct the comprehensive requirement set HML′, as illustrated in Figures 3 and 4.

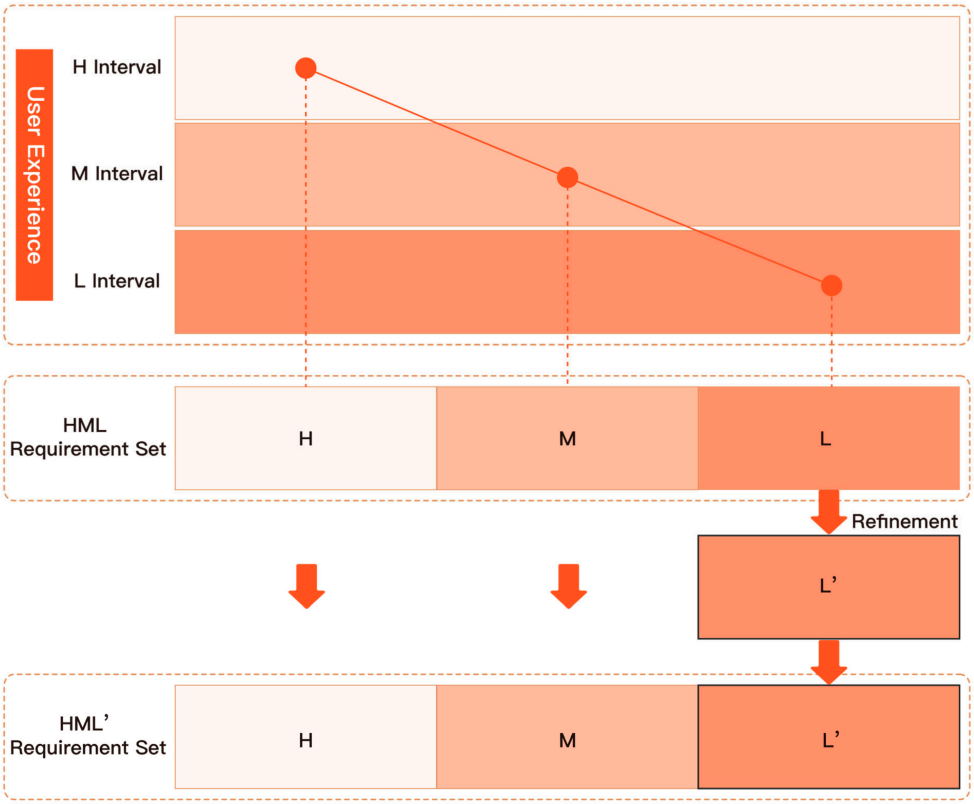


Figure 3. User experience map acquisition model for user requirement sets.

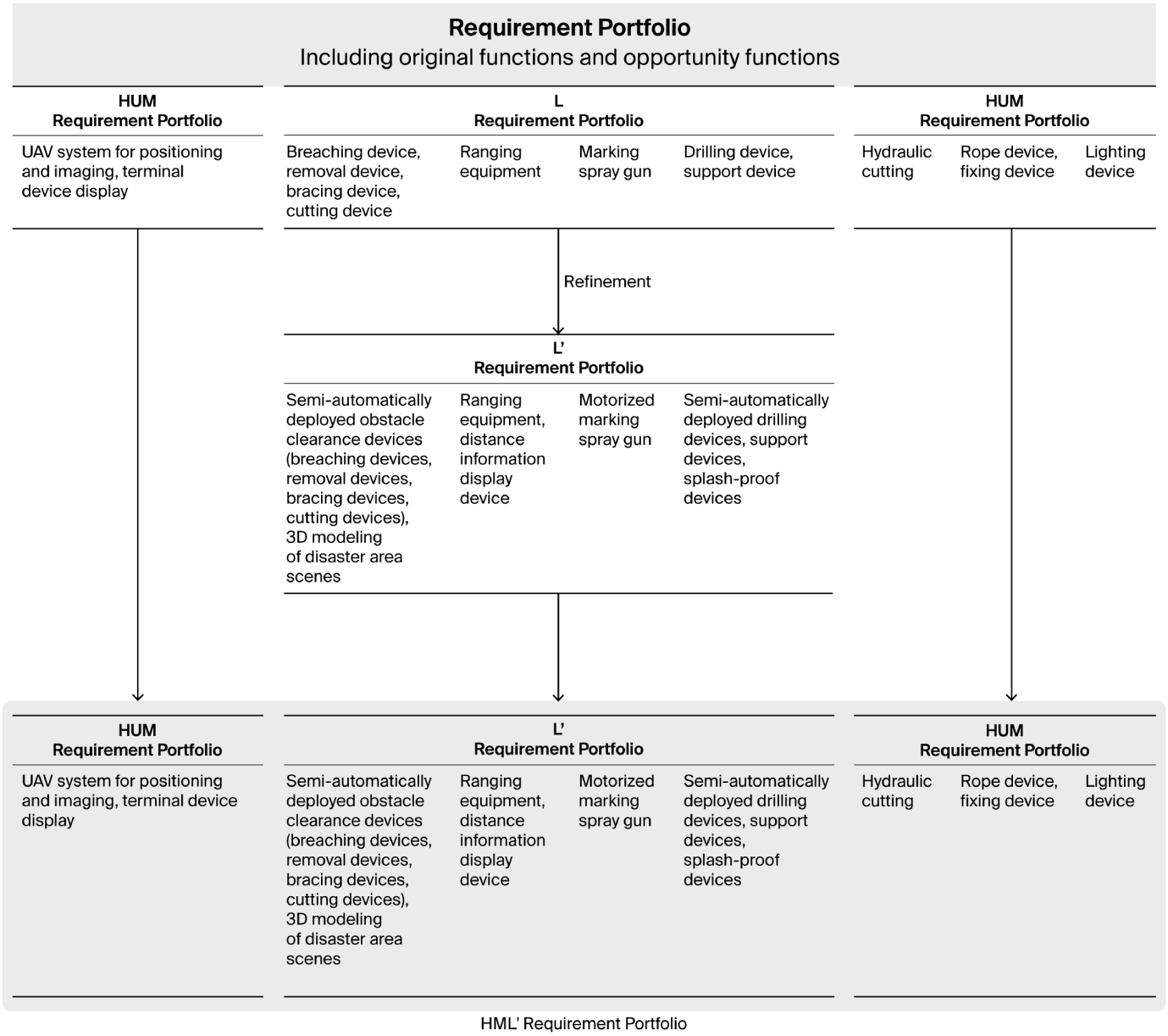


Figure 4. Demand collection based on user experience maps HML′.

Through the aforementioned methodology, 14 key user requirements (G1–G14) were ultimately distilled and categorised in detail by type and function, as shown in Table 3.

To reduce subjectivity and maintain consistency in translating UXM insights into Kano items, each preliminary requirement was linked to specific UXM evidence, such as high-load nodes, high-frequency touchpoints, negative experience fluctuations, risk perceptions, or task interruptions. The requirements were cross-checked with behavioural observation, verbal recall, NASA-TLX workload results, and expert interview records. Similar items were merged only when they referred to the same task objective or operational problem, and the final HML′ requirement set was reviewed by the research team and domain experts.

In summary, the HML′ requirement set was derived from traceable UXM evidence and multi-source verification rather than subjective interpretation alone. This process transformed experiential pain points and operational burden issues into structured requirement inputs, providing a consistent basis for subsequent Kano analysis and QFD engineering characteristic mapping.

Table 3. Categorisation and detailed description of earthquake debris removal rescue robot requirement set HML′.

No.	Requirement Category	User Requirement	Requirement Details
G1	Spatial Information Acquisition and Presentation Requirements	Positioning and imaging	Enables precise pinpointing of specific locations requiring rescue
G2		Features a logical information hierarchy and terminal display	Rescue information data must be clearly displayed, requiring terminal devices for presentation
G3		Three-dimensional reconstruction of disaster zone environments	Three-dimensional reconstruction of the disaster area environment
G4	Semi-Automated Demolition Module	Semi-automated deployment for obstacle clearance	Manoeuvrable clearance of obstacles in demolition zones in accordance with operational intent
G5		Semi-automatic deployment shearing	Performing lightweight cutting of reinforcing bars, metal pipes, timber, etc., within structures
G6		Semi-automatic deployment for drilling	Chiselling through wall obstructions to facilitate rescue personnel entry
G7		Semi-automatically deployed hydraulic cutting	Heavy-duty cutting
G8		Semi-automatically deployed supports	Equipment supports the obstacle to facilitate subsequent rescue operations
G9	Safety Protection and Visual Assurance	Splash protection	Prevents injury or impaired visibility from debris spray during operations
G10		Securing device	Ensuring personnel safety, securing straps for submerging into rooms
G11		Lighting apparatus	Lighting apparatus for rescue personnel in dimly lit spaces
G12	Range Measurement, Cognitive Assistance, and Marking Functions	Range measurement	Standardised determination of breaching perimeter
G13		Distance display	Numerical display of demolition distance
G14		Motorised marking spray gun	Mark demolition zones for easy identification by rescue personnel

5. Kano-Based User Requirements Analysis

5.1. Survey Design and Data Collection

User requirements in emergency equipment scenarios exhibit significant multidimensionality and heterogeneity, with differing mechanisms influencing user satisfaction. To identify the role of distinct requirements in satisfaction formation, this study designed a Kano questionnaire based on 14 extracted key requirements (G1–G14), incorporating items of both functional (feature presence) and reverse (feature absence) types [30]. All items employed a five-point response format (like, expect, indifferent, tolerate, dislike), with demand attribute categories determined according to the Kano standard evaluation matrix, as shown in Table 4.

This study distributed 120 questionnaires, structured as shown in Table 5, and collected 100 valid responses, yielding a response rate of 83.3%. Respondents comprised 42 emergency rescue personnel, 23 firefighting specialists, 18 equipment development engineers, and 17 users with operational experience of emergency equipment, with

2–20 years of relevant professional experience (mean = 8.6 years). Each of the 14 requirements received 100 valid paired evaluations, providing sufficient categorical frequency data for identifying the dominant Kano attribute. To check response robustness, the R and Q categories were examined; no Q responses were observed, and the average proportion of R responses was 6.93%, suggesting that the questionnaire items were generally understood consistently. Therefore, the data were considered sufficiently stable for Kano classification and subsequent QFD analysis. Informed consent was obtained from all respondents, and the survey was conducted in accordance with the approved ethics protocol. No participant overlap existed between the UXM experiential study (n = 12) and the Kano questionnaire respondents (n = 100).

Table 4. Kano model evaluation results classification comparison table.

Function/Service		Negative Question				
		Dislike (1 Point)	**Tolerable (2 Points)**	**Indifferent (3 Points)**	**As It Ought to Be (4 Points)**	**Like (5 Points)**
Positive question	Do not like (1 point)	Q	R	R	R	R
	Can tolerate (2 points)	M	I	I	I	R
	It does not matter (3 points)	M	I	I	I	R
	As it ought to be (4 points)	M	I	I	I	R
	Like (5 points)	O	A	A	A	Q

Notes. A: Attractive attribute, indicating functions that improve operational experience but do not cause strong dissatisfaction when absent; O: one-dimensional attribute, indicating performance-related functions whose fulfilment directly increases satisfaction; M: must-be attribute, indicating basic safety and reliability functions expected by rescue personnel; I: indifferent attribute, indicating functions with limited influence on satisfaction; R: reverse attribute; Q: questionable attribute. Respondents evaluated each requirement through paired functional and dysfunctional questions, and the final category was determined using the Kano evaluation matrix in Table 4.

Table 5. Example of Kano questionnaire structure design.

Requirement ID	Requirement Description	Functional Question (Perception When Functionality Exists)	Reverse-Scoring Questions (Experience When Feature Is Absent)	Response Options
G1	Positioning imaging	If the system possessed this capability, what would your feelings be?	If the system lacks this feature, how would you feel?	Like Expect Neutral Can live with it Dislike

5.2. Kano Classification Results and User Satisfaction Analysis

Based on the questionnaire's functional–reverse response design, this study categorised 14 user requirements (G1–G14) using the Kano evaluation matrix. Satisfaction coefficients (better) and dissatisfaction coefficients (worse) were calculated to quantify each requirement's contribution to user satisfaction. The better/worse coefficient formulae are as follows:

$$B = \frac{A+O}{A+O+M+I} \quad (1)$$

$$W = -\frac{O+M}{A+O+M+I} \quad (2)$$

where A, O, M, and I denote the numbers of responses classified as attractive, one-dimensional, must-be, and indifferent attributes, respectively; R and Q responses were excluded from the calculation. The better coefficient was calculated as

$(A + O)/(A + O + M + I)$, reflecting the satisfaction increase when a requirement is fulfilled, while the worse coefficient was calculated as $-(O + M)/(A + O + M + I)$, reflecting dissatisfaction when it is absent. For example, for G1 (positioning imaging), $A = 8$, $O = 11$, $M = 52$, and $I = 22$; therefore, better = $(8 + 11)/93 = 0.2043$ and worse = $-(11 + 52)/93 = -0.6774$. Since M had the highest response frequency, G1 was classified as a must-be requirement.

Based on Kano analysis of 100 valid samples, Table 6 presents the attribute classifications and satisfaction coefficient results for each requirement item. Overall, the requirements for earthquake rescue robots exhibit a clear hierarchical structure.

Table 6. Classification of earthquake rescue robot user requirement items.

No.	User Requirement	Number of Attribute Requirements						Kano Attribute	Better Coefficient	Worse Coefficient
		A	O	M	I	R	Q			
G1	Positioning imaging	8	11	52	22	7	0	Must-be (M)	0.2043	−0.6774
G2	Logically structured information hierarchy with terminal display	6	36	20	30	8	0	One-dimensional (O)	0.4565	−0.6087
G3	Three-dimensional reconstruction of disaster area scenes	54	9	8	22	7	0	Attractive (A)	0.6774	−0.1828
G4	Semi-automatic deployment clearance	20	44	9	20	7	0	One-dimensional (O)	0.6882	−0.5699
G5	Semi-automatic deployment shear	14	46	12	20	8	0	One-dimensional (O)	0.6522	−0.6304
G6	Semi-automatic deployment drilling	13	49	9	20	9	0	One-dimensional (O)	0.6813	−0.6374
G7	Semi-automatic deployment hydraulic cutting	28	39	6	20	7	0	One-dimensional (O)	0.7204	−0.4839
G8	Support for semi-automated deployment	16	43	11	21	9	0	One-dimensional (O)	0.6484	−0.5934
G9	Splash-proof	5	9	57	24	5	0	Must-be (M)	0.1474	−0.6947
G10	Fixture	10	19	49	18	4	0	Must-be (M)	0.3021	−0.7083
G11	Lighting fixture	12	8	44	28	8	0	Must-be (M)	0.2174	−0.5652
G12	Range	19	46	12	21	2	0	One-dimensional (O)	0.6633	−0.5918
G13	Distance display	8	15	57	14	6	0	Must-be (M)	0.2447	−0.7660
G14	Motorised marking gun	38	12	5	35	10	0	Attractive (A)	0.5556	−0.1889

Firstly, must-be requirements (M) primarily focus on functions directly related to safety and fundamental operational reliability, including positioning imaging, splash resistance, mounting fixtures, illumination, and distance display. These requirements exhibit significant negative sensitivity, meaning that their absence leads to marked dissatisfaction and they should be treated as basic design prerequisites. Secondly, one-dimensional requirements (O) encompass functions closely linked to performance enhancement, such as semi-automatic deployment modules, distance measurement, and information display. User satisfaction with these functions exhibits a linear response, with their better coefficients consistently high. This indicates that performance optimisation directly translates into enhanced user experience, categorising them as efficiency-oriented requirements. Furthermore, attractive requirements (A) primarily include innovative functions such as

three-dimensional scene construction and motorised marking spray guns. Satisfying these yields significant positive experiences, while their absence does not provoke dissatisfaction, exhibiting typical innovation-gain characteristics.

Collectively, these results form a multi-layered structure corresponding to must-be, one-dimensional, and attractive requirements, reflecting the progressive demand logic in rescue operations: from basic reliability to operational efficiency and scenario innovation. This categorisation will further inform weight allocation and solution optimisation within QFD.

5.3. Defining Design Objectives

The better and worse coefficients were used to support feature prioritisation under limited design resources, such as volume, weight, power supply, and module space. Requirements with more negative worse values, especially must-be requirements, were prioritised as basic safety and reliability functions, while requirements with higher better values, mainly one-dimensional or attractive requirements, were regarded as functions that could improve operational efficiency, situational awareness, or user experience. Thus, better and worse served as decision-support indicators for prioritising rescue-robot functions.

Based on the better/worse coefficients derived from the Kano model, and employing better = 0.49 and $|\text{worse}| = 0.57$ as threshold values to construct a four-quadrant distribution chart, this study has reclassified the priority levels of 14 requirements.

The first quadrant (high better, high worse) focuses on critical performance requirements that significantly enhance satisfaction and whose absence would cause marked dissatisfaction. These include demolition modules such as semi-automatic obstacle clearance, cutting, drilling, and support (G4, G5, G6, G8) alongside distance measurement functionality (G12). Consequently, establishing a continuous, high-autonomy semi-automatic demolition execution chain is designated as the primary design objective.

The second quadrant (high better, low worse) encompasses features that offer substantial gains in task transparency and decision support but are not critical, such as 3D scene reconstruction, marking spray guns, and semi-automatic hydraulic cutting (G3, G7, G14). These secondary objectives enhance situational awareness and operational fluidity.

The third quadrant (low better, low worse) encompasses items with minimal impact on satisfaction or dissatisfaction, including lighting devices (G11). These should be addressed as post-optimisation enhancements, implemented only after critical functions and safety assurances are met.

The fourth quadrant (low better, high worse) represents must-be safety and reliability requirements that should be guaranteed first. These include positioning imaging, information hierarchy and terminal display, splash protection, mounting fixtures, and distance display (G1, G2, G9, G10, G13). Their absence would significantly reduce equipment usability.

In summary, design objectives should progress according to the priority hierarchy: 'Critical Demolition Performance–Scenario Enhancement–Post-Optimisation–Fundamental Assurance'. This provides a clear basis for subsequent engineering characteristic derivation through QFD. As shown in Figure 5, the distribution of scatter points of the Kano superiority-inferiority coefficient is presented.

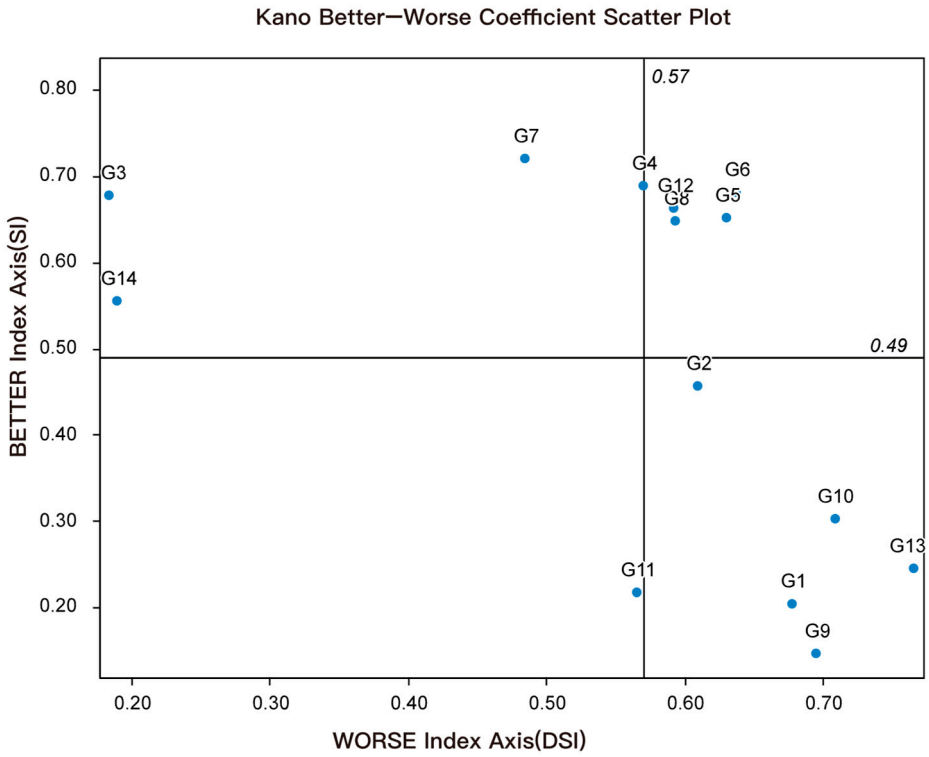


Figure 5. Kano better–worse coefficient scatter plot.

6. Design Requirement Mapping and Decision Analysis Based on QFD

In this study, QFD serves not only as a demand mapping tool but also as a mechanism for resource allocation and design prioritisation. Through QFD, the demand weights derived from Kano analysis can be further translated into technical focus areas at the design level, thereby providing quantitative justification for subsequent structural schemes and system configurations [31].

6.1. Construction of Integrated Kano–QFD Metrics and Parameter Incorporation

The integration of Kano and QFD supports the translation of user requirements into technical specifications. The Kano results were not directly used as final QFD weights. Instead, the better and worse coefficients were first converted into the customer satisfaction coefficient Cs by taking their maximum absolute value, as shown in Equation (3). The Kano attribute type was then adjusted using coefficient k in Table 7. By combining Cs, k, and the improvement ratio between current and target performance, the normalised requirement weights were obtained and input into the QFD relationship matrix to calculate the final technical priorities.

$$CS_i = \max(|Better_i|, |Worse_i|) \quad (3)$$

Table 7. Adjustment coefficient k values.

Requirement Attribute	k Value
Attractive (A)	1.5
One-dimensional (O)	1.0
Must-be (M)	0.5
Indifferent (I)	0

During the market competitiveness assessment phase, a five-point Likert scale was employed to evaluate the 14 requirements within the comprehensive demand set HML′.

The evaluation encompassed two categories: the degree to which existing earthquake demolition and rescue equipment meets actual requirements, serving as the market competitiveness indicator (Im); and the anticipated degree of requirement fulfilment based on the novel earthquake demolition and rescue robot solution proposed herein, serving as the target competitiveness indicator (It). To balance practical usage experience with professional design judgement, 18 rescue personnel and four product design experts (each with more than 5 years of relevant experience) were invited to participate in the questionnaire assessment, yielding 22 valid responses. In addition, the correlation matrix within the quality house was independently completed by three senior experts (two rescue operations specialists and one industrial design researcher). The relationship scores (1, 5, and 9) were first assigned independently and were then reviewed through a consensus discussion to resolve discrepancies and obtain the final matrix. This procedure was adopted to improve the transparency and consistency of expert judgement in the QFD mapping process. The relevant calculation methodology is illustrated in Equation (4).

$$IR_i = \frac{I_{ti}}{I_{mi}} \tag{4}$$

A higher value of IR_i indicates a greater gap between current performance and target expectation, implying a higher priority for design improvement.

To further integrate Kano attributes and design emphasis, adjustment coefficients are assigned according to requirement categories, as shown in Table 7. Attractive requirements (A) are assigned k = 1.5, one-dimensional requirements (O) k = 1.0, and must-be requirements (M) k = 0.5, reflecting their different impacts on user satisfaction. The demand influence degree is then calculated using Equation (5).

On this basis, by integrating the demand improvement rate and adjustment coefficient, the demand influence degree, considering Kano attributes and market improvement effects, is obtained, as shown in Equation (5).

$$AI_i = CS_i \cdot k_i \cdot IR_i \tag{5}$$

Further calculations yield the importance of user and market final demand, as shown in Equation (6).

$$W_i = \frac{AI_i}{\sum_{i=1}^{n} LAI_i} \tag{6}$$

Finally, the scores are input into the quality house model, with the correlation coefficient Mij assigned to three levels as 1, 5, and 9, representing a weak relationship, a moderate influence, and a very close relationship, respectively. The product is summed to yield the final importance score, which integrates user requirements, the improvement rate in market competitiveness, and design specifications, as per Equation (7).

$$H_j = \sum_{i=1}^{n} W_i \cdot M_{ij} \tag{7}$$

Building on this, the satisfaction parameters derived from the Kano model were integrated with market competitiveness indicators to perform a comprehensive calculation of requirement importance using Equations (3)–(6). All quantitative results presented in Table 8 were recalculated to ensure strict consistency with the defined formulations. This establishes a continuous and traceable quantitative process encompassing user experience acquisition, requirement classification, competitiveness evaluation, and final determination of requirement weights, thereby providing a reliable basis for subsequent quality house analysis.

Table 8. QFD numerical results based on requirement set HML′.

Requirement ID	Requirement Details	CS_i	k_i	I_t	I_m	IR_i	AI_i	W_i
G1	Positioning imaging	0.677	0.5	3.50	2.32	1.51	0.51	0.055
G2	Logically structured information hierarchy and terminal display	0.608	1.0	3.73	1.27	2.94	1.79	0.192
G3	Three-dimensional reconstruction of disaster area scenes	0.677	1.5	3.45	1.32	2.61	2.65	0.285
G4	Semi-automatic deployment clearance	0.688	1.0	4.73	4.09	1.16	0.80	0.086
G5	Semi-automatic deployment shearing	0.652	1.0	4.18	3.77	1.11	0.72	0.078
G6	Semi-automatic deployment drilling	0.681	1.0	4.23	3.95	1.07	0.73	0.078
G7	Semi-automatic hydraulic cutting	0.720	1.0	4.45	3.91	1.14	0.82	0.088
G8	Support for semi-automated deployment	0.648	1.0	4.05	3.45	1.17	0.76	0.082
G9	Splash-proof	0.695	0.5	4.05	1.45	2.79	0.97	0.104
G10	Fixture	0.708	0.5	4.27	3.27	1.31	0.46	0.049
G11	Lighting fixture	0.565	0.5	3.68	2.36	1.56	0.44	0.047
G12	Range	0.663	1.0	3.55	1.45	2.45	1.62	0.174
G13	Distance display	0.766	0.5	3.59	1.59	2.26	0.87	0.093
G14	Motorised marking gun	0.556	1.5	3.59	1.59	2.26	1.89	0.203

Note: IR_i denotes the improvement ratio of user requirements; CS_i represents the customer satisfaction coefficient derived from the Kano model; k_i is the Kano attribute adjustment coefficient, used to regulate the relative influence of different requirement types; AI_i denotes the adjusted importance; and W_i represents the final weight, used to determine the priority of technical features in the QFD analysis.

6.2. Design Requirement System Construction

Once the final priority of user requirements is determined, it is necessary to further translate abstract user-level needs into executable engineering design specifications. In accordance with GB/T 29178-2012, fire emergency rescue guidelines for equipment allocation, issued by the General Administration of Quality Supervision, Inspection and Quarantine of the People's Republic of China and the Standardization Administration of China, the configuration of earthquake rescue equipment should follow the principles of specificity, compatibility, complementarity, efficiency, and coordination [32].

Integrating the operational characteristics of earthquake demolition rescue robots, engineering constraints, and user pain points, this study systematises system-level design objectives to establish a multi-tiered design requirement framework. Primary-level design requirements define the overall design direction, forming five core principles centred on specificity, compatibility, efficiency, complementarity, and user-friendliness. Building upon this foundation, specific design elements are further refined with engineering feasibility as the guiding principle, resulting in corresponding secondary design requirements that are systematically numbered.

This design requirement framework ensures logical integrity while providing a clear and stable structural basis for quality house mapping [33]. It is incorporated into the quality house calculation process as the final design requirements, as illustrated in Table 9 [33].

Table 9. Design requirement indicator details.

Serial No.	Primary Design Requirement	Description	Secondary Design Requirement	Request Number
a1	Relevance	The configuration of rescue equipment shall be determined based on the nature of rescue missions and typical disaster scenarios, while comprehensively considering local natural conditions and economic development levels. Equipment performance must meet fire emergency rescue requirements.	Modular design adaptable to diverse disaster scenarios	D11
			Flexibility for adaptation to diverse geographical environments	D12
a2	Compatibility	The configuration of rescue equipment must ensure system compatibility, rational pairing, comprehensive functionality, and adequate quantity.	Integrated design of system compatibility	D21
			Functionally diverse configuration	D22

Table 9. *Cont.*

Serial No.	Primary Design Requirement	Description	Secondary Design Requirement	Request Number
a3	Efficiency	Priority should be given to selecting equipment that is superior in performance, lightweight and efficient, versatile in function, and highly adaptable. Regular inspections and assessments should be conducted to evaluate the effectiveness and operational efficiency of deployed equipment, promptly phasing out outdated or low-performance items.	Versatile and highly adaptable systems	D31
a4	Complementarity	Urban search and rescue equipment configurations should possess functional complementarity and on-site substitutability. Should one type of equipment become damaged, its functions should be achievable using other equipment.	Regular inspection and maintenance design	D41
a5	User-friendliness	Rescue equipment should provide a seamless user experience and comply with human–machine data specifications.	Seamless user operation experience	D51
			Intuitive user interface design	D52
			Human-centred design equipment	D53

6.3. Quality House Mapping and Design Requirement Weighting Calculation

During the quality house mapping process, practitioners with hands-on experience in earthquake rescue demolition operations and experts in product design were invited to assess the degree of correlation between user requirements and design specifications. Correlation strength was quantified using three levels as 1, 5, and 9, representing weak, moderate, and strong correlations, respectively, and denoted by symbols ○, ●, and ▲.

The design requirement weighting results are presented in Table 10. D53 (human-centred, considerate equipment provision), D21 (system-integrated design), and D22 (comprehensive, versatile configuration) ranked highest among all design requirements. This ranking was considered relatively stable because these items showed clear score advantages in the quality house and were supported by multiple high-priority user requirements rather than a single mapping judgement. Therefore, they were identified as the key technical priorities for subsequent design development.

Table 10. Relationship matrix between seismic rescue breaking segments and design requirements.

Requirement	W_i	D11	D12	D21	D22	D31	D41	D51	D52	D53
G1	0.055	○	●			○		○	○	
G2	0.192			●			○	▲	▲	▲
G3	0.285		○			○		○	○	
G4	0.086	○	●	▲	●	●				
G5	0.078	●	●	▲	●	●				
G6	0.078	●	●	▲	●	●				
G7	0.088	●	●	▲	▲	●				
G8	0.082	●		▲	▲					
G9	0.104			▲	▲					▲
G10	0.049		●			●				
G11	0.047		●	●						▲
G12	0.174	○	○							▲
G13	0.093	○			●	●		●		▲
G14	0.203		○		●	●		●		●
Score		2.038	3.067	5.839	5.156	3.715	0.192	3.548	2.068	6.505
Sort		8	6	2	3	4	9	5	7	1

Note: In the table, ▲ denotes strong correlation, ● denotes moderate correlation, ○ denotes weak correlation, and blank cells indicate no correlation.

In summary, the design of earthquake rescue robots should prioritise human-centred safety and care configurations, system-level integration, and functional diversity and synergy. This design guidance provides a clear basis for subsequent functional definition

and structural mapping using the FBS model, helping ensure alignment between the design solution and actual rescue scenarios and genuine user requirements.

7. Conceptual Design Derivation Based on FBS

The function–behaviour–structure (FBS) model provides a systematic framework for interpreting the transformation of design intent into product form by distinguishing functional objectives, enabling behaviours, and physical structures. Given the complex functionality, high structural coupling, and requirement for traceable design decisions in seismic demolition rescue robots, this paper introduces the FBS model as the key methodology to integrate UXM–Kano–QFD analysis outcomes. By incorporating the weightings of design requirements derived from QFD into the FBS framework, these serve as input criteria for the functional layer (F). High-weight requirements translate into core functional constraints, and medium-weight requirements guide performance trade-offs at the behavioural layer (B). In contrast, low-weight requirements primarily influence form and detail optimisation at the structural layer (S). This achieves a systematic translation of quantitative requirement weightings into functional–behavioural–structural decisions.

Integrating user experience insights, requirements analysis, and QFD-derived design specifications, a functional–behavioural–structural mapping for the seismic demolition rescue robot is established. Its overall structure is illustrated in Figure 6.

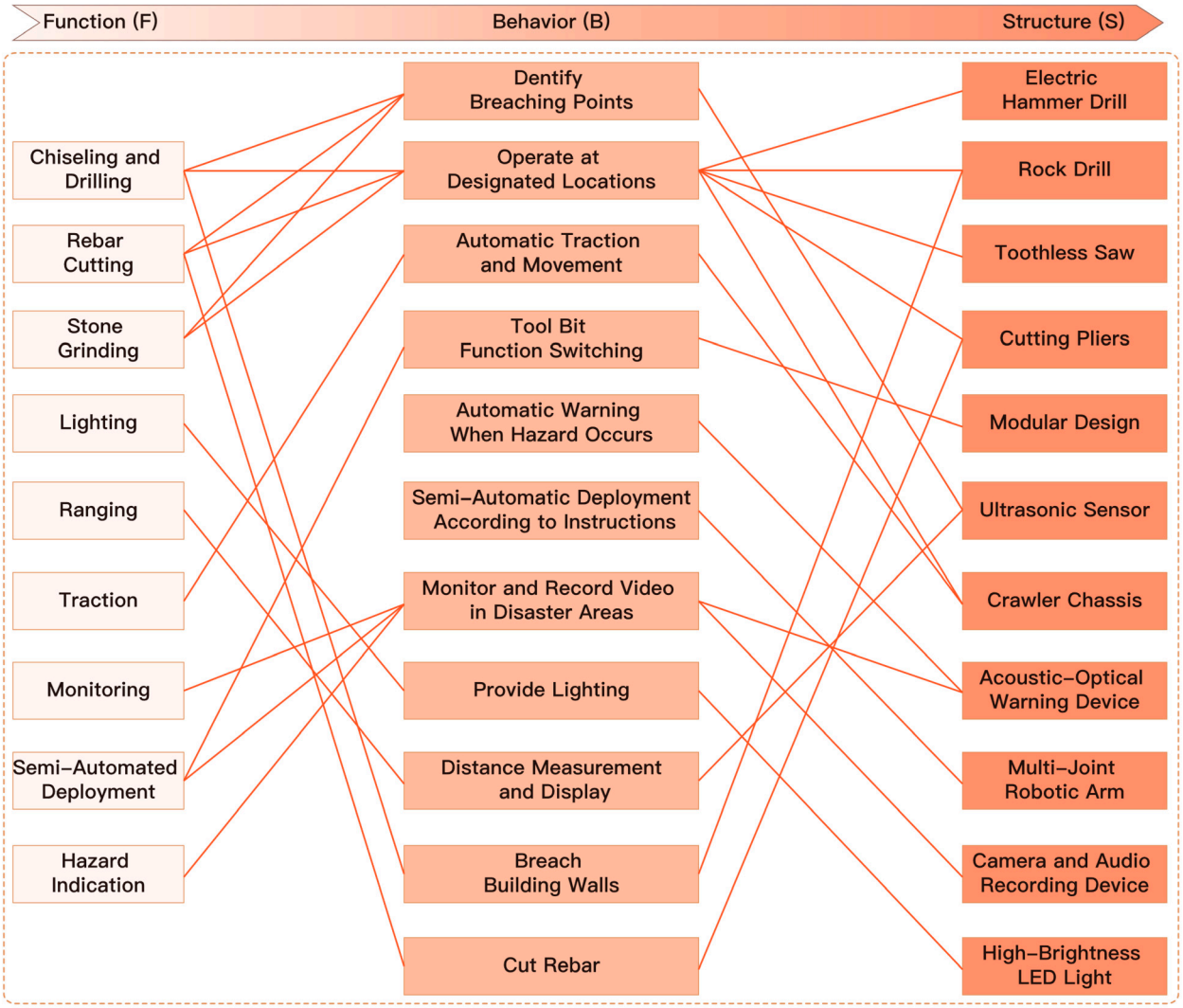


Figure 6. Functional–behavioural–structure mapping diagram for earthquake rescue robots.

In summary, the function–behaviour–structure (FBS) mapping is not conducted as a form-driven design activity but instead represents the structured translation of high-priority technical requirements identified through Quality Function Deployment (QFD) analysis. Specifically, technical features with higher design requirement weightings, such as D53 and D21, are explicitly mapped to corresponding functional constraints and structural strategies. This ensures that conceptual design decisions are directly traceable to quantified user and system requirements.

7.1. Function–Behaviour Mapping

At the functional-behaviour mapping level, the drilling function is refined to execute drilling operations at designated demolition points, enabling localised removal of concrete and obstructions. The rebar cutting function is specified to perform targeted shearing operations upon identifying reinforcing bars or metallic components. The stone grinding function involves continuous abrasion of rubble or concrete edges to clear the demolition area. The illumination function is refined to provide continuous task lighting during demolition and detection, ensuring operational visibility in low-visibility environments. The distance measurement function is specified to measure distances between demolition targets and surrounding structures, providing quantitative basis for the operational scope [34]. The indexing function is detailed to mark key locations within the demolition area, aiding rescue personnel in rapid positioning.

Regarding monitoring and safety, the monitoring function entails continuous observation and video recording of demolition processes. Hazard alerting is refined to automatically signal risk information when potential dangers or abnormal conditions are detected.

Furthermore, the semi-automated deployment function is specified to automatically guide mechanical component movements according to operational instructions, whilst executing functional switching of cutting heads or tools to adapt to diverse demolition tasks [35].

7.2. Behaviour–Structure Mapping and Structural Strategies

At the behaviour-structure mapping level, the aforementioned operational actions are further translated into demolition execution structures such as hammer drills, rock drills, toothless saws, and grinding cutters, alongside perception and display structures including ultrasonic sensors, cameras, audible–visual warning devices, and high-intensity LED lights.

To fulfil the operational requirement of demolition at designated locations, automated traction and mobility mechanisms are configured to enable precise positioning of tools at target demolition points. For rapid task switching during demolition operations, interchangeable cutting heads and a modular design are employed, allowing drilling, cutting, and grinding to be performed on a single platform [36].

For distance measurement and monitoring, integrated ultrasonic sensors and camera systems perform distance measurement, environmental observation, and task documentation. Hazard alerting incorporates audible–visual warning devices and high-intensity LED lights to provide immediate indications of potential risks.

These structural strategies establish a clear correspondence between the demolition rescue robot's functional objectives, operational behaviours, and physical architecture, providing explicit guidance for subsequent system integration and detailed design.

7.3. Conceptual Design Derivation Process

To control conceptual complexity while maintaining traceability in requirement translation, this study prioritised the three highest-ranked design requirements identified by

the QFD analysis—D53, D21, and D22 with weighted scores of 6.505, 5.839, and 5.156, respectively—as the core inputs for FBS derivation, while the remaining requirements were incorporated as secondary constraints.

As shown in Table 10, based on the prioritisation of design requirements, the key design focuses for the earthquake rescue demolition phase are, in order of importance: human-centred equipment configuration (D53), system integration design (D21), and functionally complete multi-purpose structural configuration (D22). Design resources are allocated accordingly. By integrating the FBS mapping relationship, the conceptual design derivation based on the FBS model has been completed.

To navigate the complex post-earthquake terrain of steep slopes littered with rubble and mud, a tracked undercarriage with superior manoeuvrability and traction was selected as the load-bearing transport system. An electric drive system, offering high energy efficiency, precision, and adaptability, was chosen for the robotic arm unit.

Meeting the D53's human-centred design requirements for considerate equipment provision, ergonomic considerations were integrated into the design practice. Audible and visual warning units were fitted to both sides of the chassis. During operations, these units delineate hazardous zones through visual and audible signals, safeguarding both the physical and psychological well-being of rescue personnel. As shown in Figure 7, an oblique front functional schematic of the earthquake demolition and rescue robot is presented.

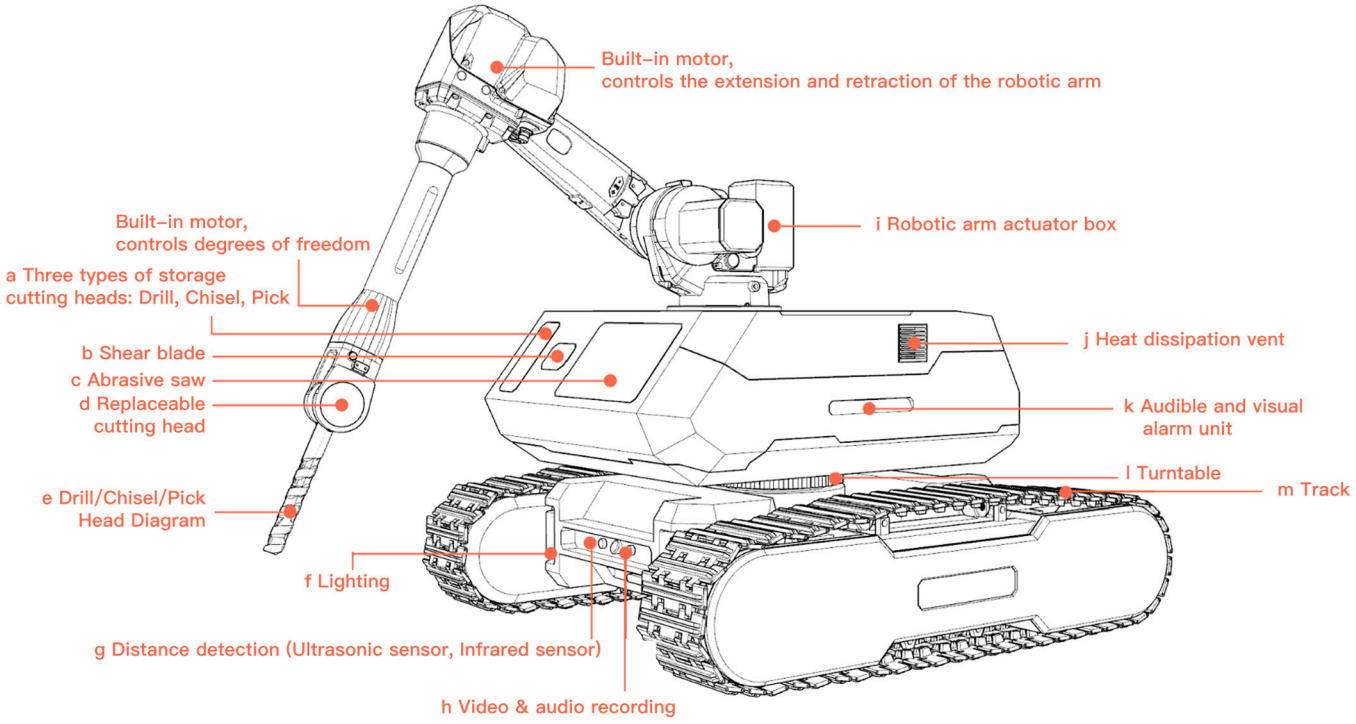


Figure 7. Earthquake demolition rescue robot—oblique front angle functional illustration.

This design effectively alerts responders to potential risks, minimising operational injuries and alleviating psychological strain, which ensures unimpeded rescue operations [37].

The D21 system features an integrated design with modular functional units mounted on the chassis' front section to accommodate diverse equipment selection requirements. These include a combined drill–chisel unit, a cutting blade unit, and a toothless saw unit, tailored to meet specific operational demands across various rescue scenarios.

The drill–chisel–pick combination blade unit connects to the end of the electric robotic arm via a quick-change mechanism, enabling operators to swiftly perform manual blade changes. Taking the drill bit of the drill–chisel–pick unit as an example, the blade change procedure is illustrated in Figure 8. The cutting blade unit interfaces with the robotic arm's end effector; when the arm, fitted with the cutting blade, is positioned at the work site, a linkage mechanism drives the cutting blade in reciprocating motion to achieve efficient cutting. To enhance the operational capacity of the toothless saw unit, an additional external motor is incorporated. Through manual deployment, the toothless saw unit can be precisely and flexibly attached to the robotic arm, enabling reinforced operations. Upon activation of the robotic arm's end-effector circuitry, the unit performs cutting functions on structural elements.

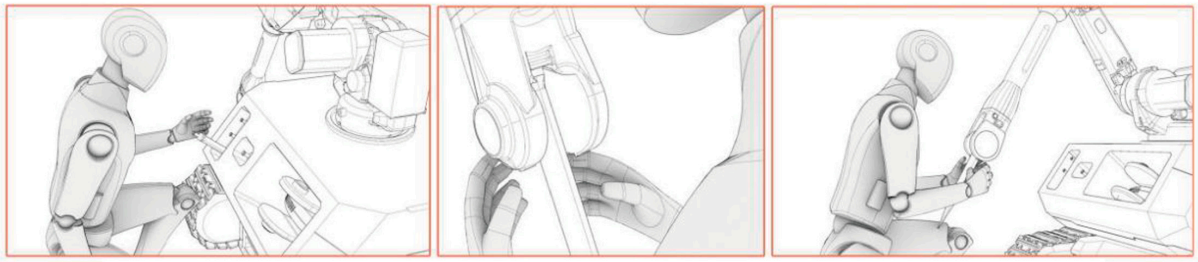


Figure 8. Blade replacement illustration.

This design not only improves the adaptability and flexibility of rescue equipment but also significantly enhances operational efficiency and response times through modularisation and rapid tool-changing mechanisms.

To meet the requirements for a fully functional and versatile configuration design for the D22, the rear compartment is equipped with manual tools including wrenches, hammers and nails, towing equipment and basic lighting, as illustrated in Figure 9.

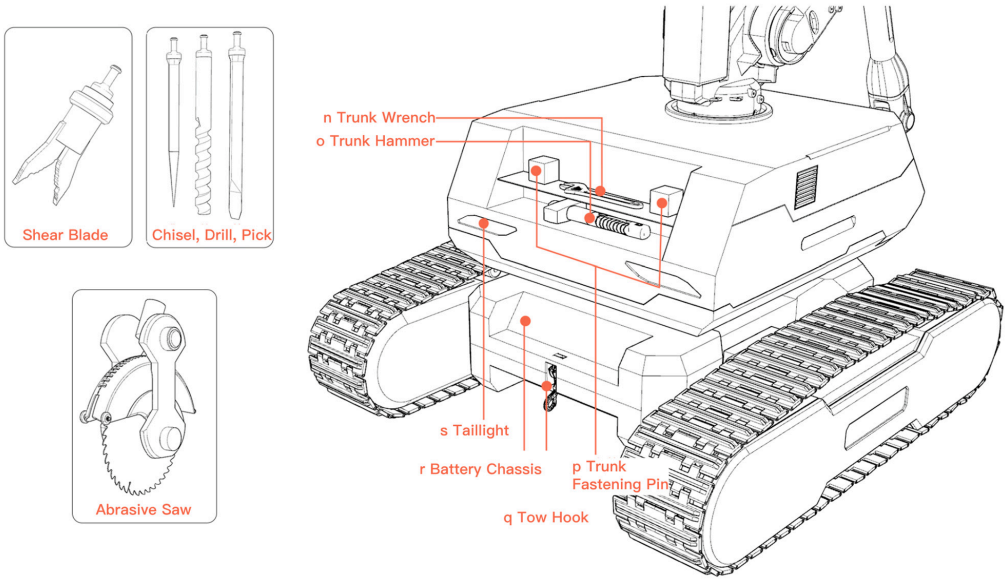


Figure 9. Earthquake demolition rescue robot—oblique rear angle functional illustration.

7.4. Design Solution Output

Earthquake rescue robots must operate in complex terrain, requiring designs that balance robustness, durability, and operational recognisability. To this end, the design phase incorporates a compact protective casing that structurally fulfils stringent operational

safety requirements (D53), thereby enhancing impact resistance and reducing exposure risks during demolition work. This conveys both strength and reliability, fostering greater trust among rescue personnel and disaster victims.

During sketch development, intricate surface transitions and complex structural elements were eliminated. The form was constructed with clean contours and minimalist straight lines and curves. Localised surface treatments were applied while preserving functional necessities. Reducing redundant surfaces and transitions, the design employs large-scale volumetric blocks complemented by minimal auxiliary cuts. This yields a clean, crisp overall form, facilitating rapid identification and operation amidst chaotic environments, as illustrated in Figure 10.

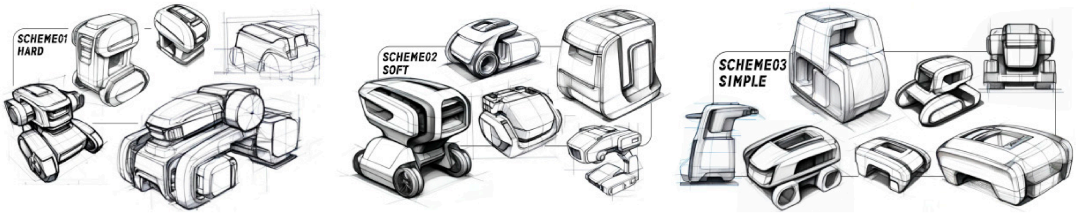


Figure 10. Conceptual sketch evolution.

Modelling was undertaken using Rhino 7, a computer-aided design software. The main body shell is based on a rectangular prism with chamfered edges, optimised to balance structural stability and sculptural layering. The chassis exhibits bilateral symmetry, with a robotic-arm turntable at the front. The front and side panels integrate cooling, tool storage, and warning functions, while the rear houses fixed and emergency lighting fixtures. Detection and radar sensors are positioned on the top. Both internal and external structural designs ensure convenient equipment maintenance and efficient heat dissipation.

Import the model into KeyShot 11 for material and lighting rendering, emphasising the safety and strength of the earthquake rescue robot. The geometric surface structure and linear form deliver a crisp visual experience. Following rendering, Photoshop 2022 was employed to refine colour and lighting. The primary body colour features a metallic finish, complemented by bright orange–yellow rescue markings as accent detailing to reinforce its operational purpose. This approach lends the overall design a functional-chassis aesthetic, reducing production and maintenance costs while directing visual focus to the robotic arm and core operational zones [38]. As illustrated in Figure 11, this visualisation demonstrates the functional layout and modular configuration.



Figure 11. Conceptual configuration visualisation of rescue robots.

8. Process-Level Simulation and Methodological Validation

To evaluate the conceptual-level rationality of the proposed earthquake demolition rescue robot design scheme based on the UXM–Kano–QFD–FBS methodology in terms of task continuity, functional coordination rationality, and overall usability, this paper conducts process-level simulation and perceived-usability evaluation.

8.1. Validation Objectives and Experimental Design

The verification objectives primarily encompass the following two aspects: (1) evaluating the design scheme for process integrity and task-flow rationality within typical demolition task chains, with particular emphasis on whether demolition tasks can form a continuous, closed-loop operational workflow; and (2) assessing the design scheme's usability and overall usability level from a user perspective in complex, high-risk demolition scenarios to examine whether the conceptual interaction logic aligns with rescue personnel's operational cognition and behavioural habits.

To achieve these objectives, this paper employs a combined validation approach that integrates objective task-flow simulation verification with subjective evaluation using the SUS (System Usability Scale) methodology.

8.2. Demolition Task Workflow Simulation Validation

Based on the preceding UXM analysis and FBS, function–behaviour–structure, mapping results, representative task sequences from the demolition phase of earthquake rescue operations were selected as simulation subjects. These encompassed critical stages including target localisation, demolition execution, spatial support, and rescue deployment. During simulation, the design scheme underwent procedural validation centred on the operational sequence, module switching, and functional integration relationships inherent to demolition tasks.

Building upon the core operational behaviours defined in the preceding FBS model, a Stateflow workflow model for the demolition rescue robot was constructed, with its overall structure illustrated in Figure 12.

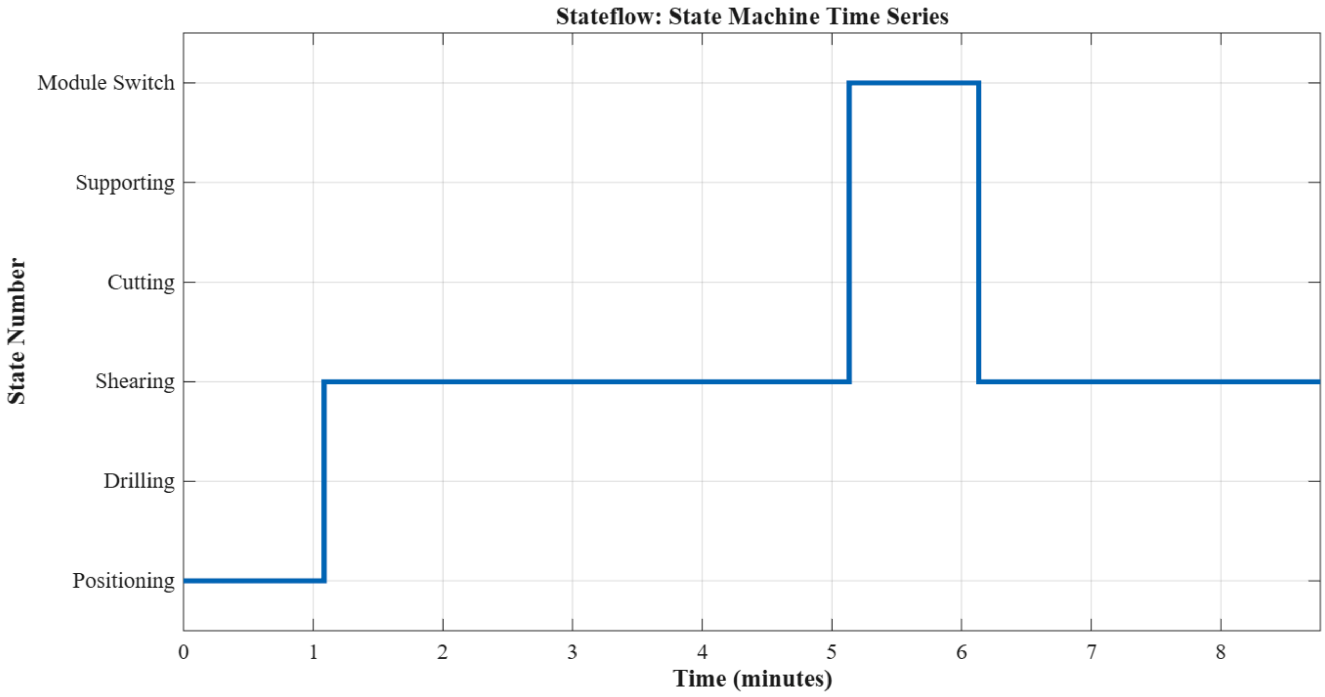


Figure 12. Overall operational process state machine.

The model organises the demolition and rescue process into key states along the task sequence, comprising S1: environmental orientation and target verification state; S2: demolition tool deployment state; S3: demolition operation execution state; S4: module switching and adjustment state; and S5: demolition completion and safety confirmation state. Transitions between states are achieved through event triggers and conditional evaluations, forming a complete task chain logic. This demonstrates the overall state machine structure of demolition rescue operations, with states arranged in sequence according to actual rescue procedures to ensure consistency between the simulation model and real-world scenarios.

Building on the overall state model, a detailed simulation analysis is conducted of typical high-frequency operational workflows in demolition rescue.

Firstly, the state transition relationships for the fundamental 'locate-demolish' workflow. Simulation results indicate that upon completing environmental perception and target confirmation, the system can smoothly transition into demolition preparation and execution states. State transition conditions are clearly defined with no logical conflicts, as illustrated in Figure 13.

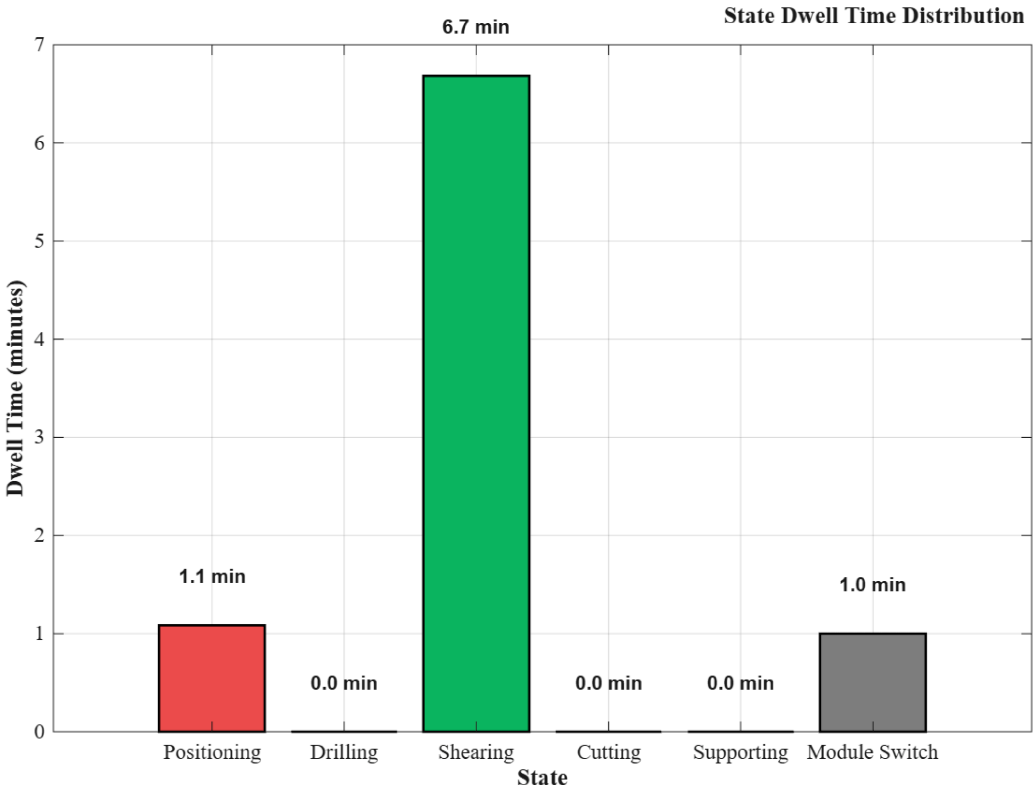


Figure 13. "Positioning–Demolition" foundational process.

Subsequently, the 'demolition—module switching—re-demolition' workflow was validated. Simulation results demonstrated that upon completing a demolition task, the system could trigger a module-switching state in accordance with operational requirements. Following the completion of the switch, it would re-enter the demolition execution state, thereby forming a continuous operational closed loop. This outcome validated the rationality of the modular demolition structure design proposed earlier at the workflow level, as illustrated in Figure 14.

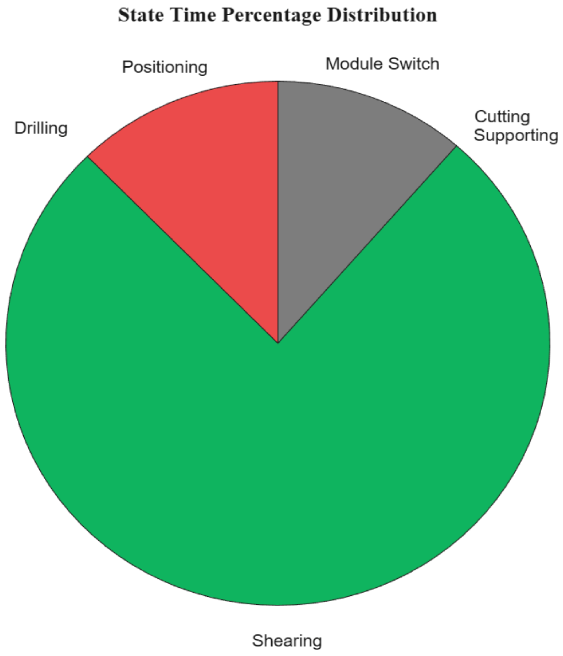


Figure 14. Module-switching process simulation.

Finally, the state response process for multiple consecutive demolition tasks was further simulated. Results demonstrated that during repeated transitions between demolition and switching states, the state machine operated stably. Each state exited normally upon satisfying its exit conditions, with no deadlocks or abnormal jumps occurring. This indicates the design possesses robust process stability, as illustrated in Figure 15.

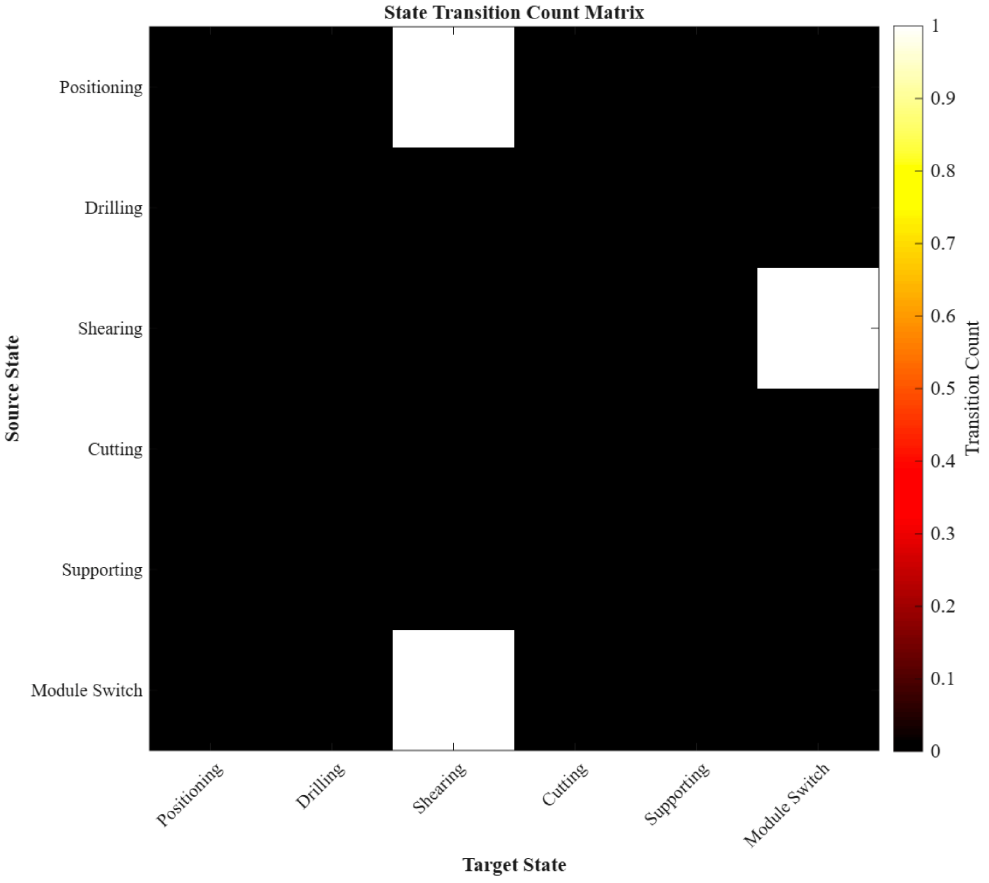


Figure 15. Continuous demolition process.

Given the stringent requirements for human–machine collaboration and safety at earthquake rescue sites, this study incorporates semi-automatic operational logic into the simulation, focusing on validating human–machine interaction nodes.

In semi-automatic control mode, the system requires manual confirmation at critical junctures, including demolition initiation, module-switching verification, and task-completion assessment. Simulation results demonstrate that state transitions remain clear and seamless in this mode, effectively mitigating the risk of erroneous triggering or switching. This indicates the conceptual rationality of the design in terms of safety logic and controllability, as illustrated in Figure 16.

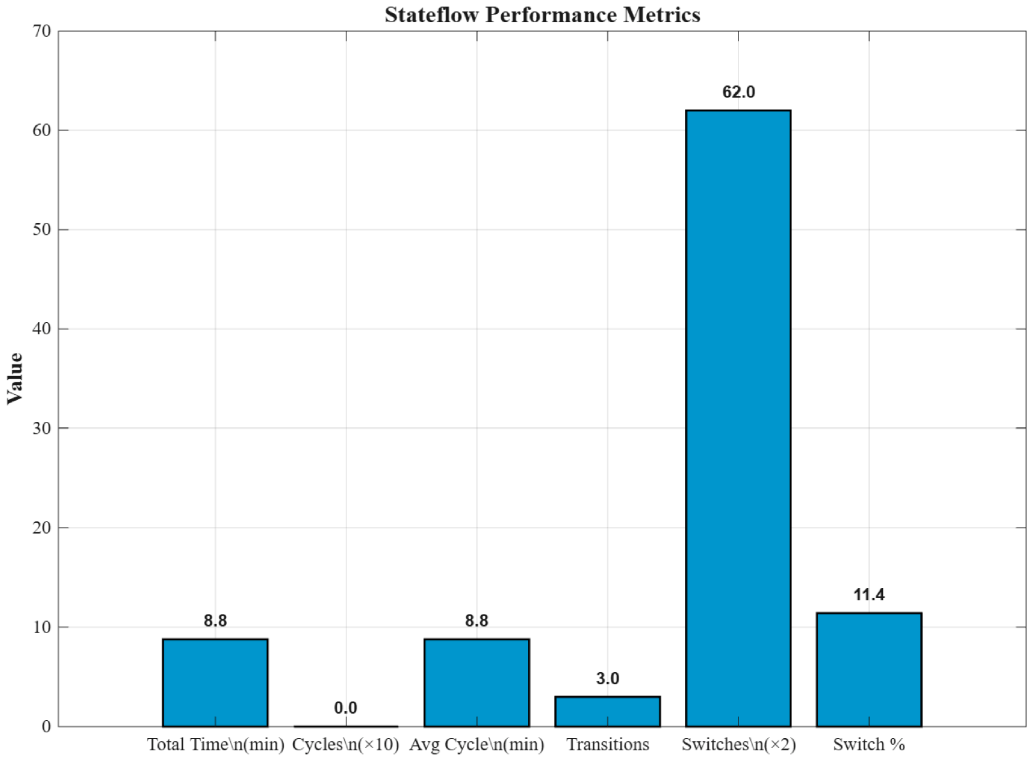


Figure 16. Semi-automatic operation mode.

Following the completion of the Stateflow workflow simulation, the simulation results were comprehensively compared with the preceding UXM–Kano–QFD–FBS analysis outcomes, as illustrated in Figure 17.

The comprehensive analysis indicates: (1) the simulation workflow exhibits high consistency with critical task nodes identified in UXMs, demonstrating the design's capacity to address high-load experience touchpoints effectively; (2) module switching and continuous operation logic within Stateflow align with high-weight design requirements (D21, D22) in QFD, supporting the traceability of requirement translation; and (3) the stable closed-loop operational workflow provides process-level validation support for the structural solution derived via FBS.

Consequently, the simulation results indicate that the proposed demolition rescue robot design possesses logical integrity and state stability at the workflow level. These results provide process-level evidence for the UXM–Kano–QFD–FBS methodology chain in modelling demolition task workflows and coordinating module design, but do not constitute field-operational validation.

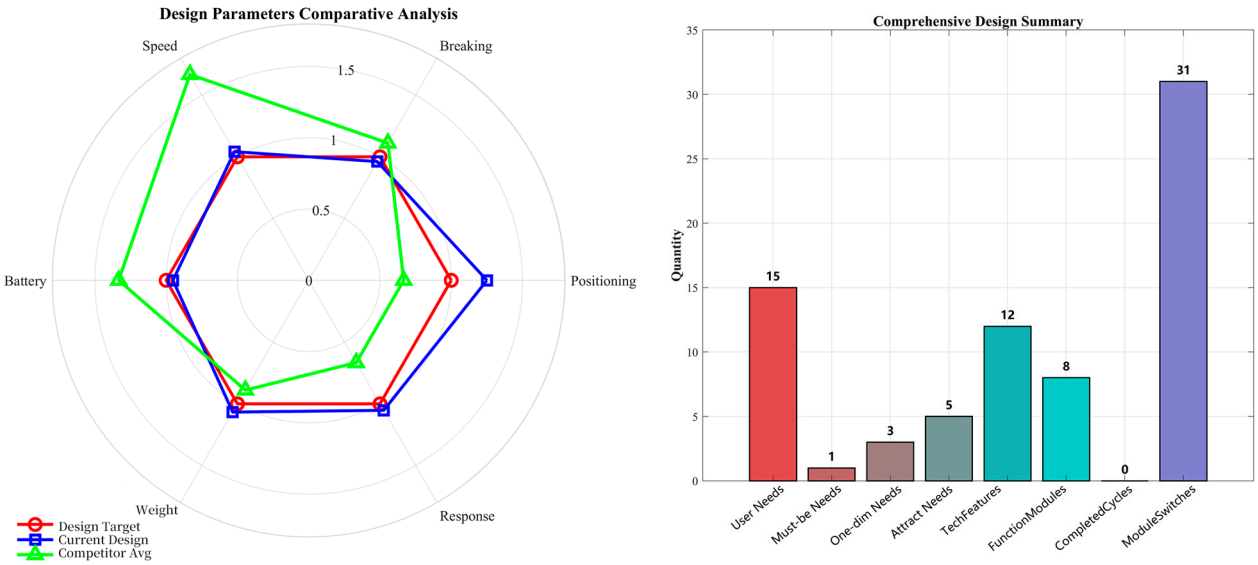


Figure 17. Comprehensive analysis and evaluation.

8.3. Usability Evaluation of Design Schemes Based on SUS

Following the completion of process-level simulation, the perceived usability of the conceptual design scheme was assessed using the System Usability Scale (SUS), as shown in Table 11.

Table 11. SUS score grading scale.

SUS Score Grades	Rating	Percentage Grade
84.1–100	A+	96–100
80.8–84	A	90–95
78.9–80.7	A−	85–89
77.2–78.8	B+	80–84
74.1–77.1	B	70–79
72.6–74	B−	65–69
71.1–72.5	C+	60–64
65–71	C	41–59
62.7–64.9	C−	35–40
51.7–62.6	D	15–34
0–51.7	F	0–14

Developed by Brooke in the mid-1980s, this scale constitutes a standardised usability assessment tool. It has been extensively validated to maintain high reliability and stability even under small-sample conditions, making it particularly suitable for engineering equipment and emergency system research in high-risk settings where large-scale subject recruitment is challenging. As a rapid standardised questionnaire for overall usability assessment, the SUS comprises 10 items and two sub-questionnaires [39]. As shown in Figure 18, the graphical usability evaluation scale of the SUS earthquake demolition and rescue robot is presented. This study uses the SUS to conduct the product usability evaluation. Scores are rated according to the industry-standard SUS letter grading norm, with the ratings from highest to lowest being A+, A, A−, B+, B, B−, C+, C, C−, D, and F. The corresponding score ranges and percentile benchmarks for each grade are detailed in Table 11.

No.	Statement	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1	I would be willing to use the earthquake rescue robot frequently.	1	2	3	4	5
2	I think the earthquake rescue robot is too complicated.	1	2	3	4	5
3	I think the earthquake rescue robot is very easy to use.	1	2	3	4	5
4	I need professional help to use the earthquake rescue robot.	1	2	3	4	5
5	I think the multiple functions of the earthquake rescue robot are well integrated.	1	2	3	4	5
6	I think the earthquake rescue robot has too many inconsistent parts.	1	2	3	4	5
7	I think most people can quickly learn to use the earthquake rescue robot.	1	2	3	4	5
8	I find the earthquake rescue robot very troublesome to use.	1	2	3	4	5
9	I feel confident when using the earthquake rescue robot.	1	2	3	4	5
10	I need to learn a lot to use the earthquake rescue robot.	1	2	3	4	5

Figure 18. Graphic usability evaluation scale of SUS for the earthquake demolition rescue robot.

A conceptual interface prototype was developed for SUS evaluation. After a brief explanation of the operating procedures and interaction logic, 23 participants (15 rescue personnel and 8 equipment design professionals; 20 male, 3 female; aged 24–48 years, mean = 33.7) completed a task-based interaction protocol and then filled in the SUS questionnaire. The protocol included five stages: environmental orientation and target confirmation, demolition tool deployment, demolition operation execution, module switching and adjustment, and demolition completion with safety confirmation. Each participant interacted with the prototype for approximately 22 min, covering familiarisation, task execution, and post-task feedback. During the process, participants were required to operate or confirm key workflow nodes, including target localisation, module selection, tool switching, task execution, safety warning recognition, and final confirmation, rather than only viewing static renderings. Informed consent was obtained from all participants. Five participants had also taken part in the earlier Kano survey, but none had participated in the UXM experiential study.

The SUS questionnaires were scored using the standard method: positive odd-numbered items were converted to $xi-1$, and negative even-numbered items to $5-xi$, where xi represents the original score. The raw SUS score was calculated by summing all item scores and multiplying by 2.5. The final mean SUS score was 80.22, corresponding to grade A−, indicating a relatively high level of perceived usability at the conceptual interaction stage. However, because the evaluation was based on a conceptual interface prototype rather than a physical robot, the result does not reflect ergonomic or mechanical performance and should not be interpreted as proof of field readiness.

In summary, the Stateflow simulation verified task-flow logic and module coordination, while the SUS evaluation reflected perceived usability at the conceptual interaction level. Under uncertain real-world disaster conditions, the proposed methodology may support requirement tracing, function prioritisation, and iterative optimisation; however,

its operational effectiveness must be further evaluated through physical prototype testing and controlled field trials.

9. Discussion

9.1. Summary of Findings

This study developed and applied an integrated UXM–Kano–QFD–FBS methodology for the design of earthquake demolition rescue robots. The UXM-based experience analysis identified critical pain points and high-load nodes across the positioning, demolition, and spatial rescue phases, providing structured input for requirement extraction. The Kano classification revealed a clear hierarchical structure among the 14 identified requirements, distinguishing must-be safety and reliability requirements, one-dimensional performance-related requirements, and attractive experience-enhancing requirements. The QFD analysis translated these prioritised requirements into weighted engineering design specifications, with human-centred equipment configuration (D53), system integration (D21), and multi-functional configuration (D22) emerging as the highest-priority design elements. The Stateflow simulation indicated logical integrity and state stability of the proposed task workflow, while the SUS evaluation yielded a score of 80.22 (Grade A−), suggesting satisfactory perceived usability at the conceptual level.

9.2. Methodological Implications

The proposed framework makes several methodological contributions. First, the integration of UXMs with Kano–QFD establishes a traceable chain from experiential insight to engineering specification, addressing the fragmentation observed in prior rescue equipment design studies [8,12]. Second, the HML′ requirement aggregation model provides a structured approach for converting complex experiential pain points into quantifiable design inputs, which may be applicable to other domains involving extreme operational environments. Third, the combination of process-level simulation (Stateflow) with subjective usability evaluation (SUS) offers a dual-validation strategy that is particularly relevant for early-stage design assessment where physical prototypes are not yet available.

9.3. Limitations

Several limitations of this study should be acknowledged. First, the UXM experiential study involved 12 male participants, reflecting the gender composition of earthquake demolition rescue teams in China. While this sample is representative of the target user population, the findings may not generalise to mixed-gender teams or rescue operations in other cultural contexts. Future studies should consider broader demographic sampling where feasible. Second, the Kano survey (n = 100) and QFD expert panel (n = 22) provided adequate data for the analytical methods employed, but larger and more diverse samples would strengthen the external validity of the requirement prioritisation. Third, the SUS evaluation was conducted using a conceptual operational interface rather than a physical prototype; consequently, the usability score reflects process-level interaction quality rather than the ergonomic or mechanical performance of a built system. Physical prototype testing in realistic operational environments constitutes an essential next step. Fourth, the quality house correlation matrix was completed by a small expert panel (n = 3) using a consensus-based approach. Although D53, D21, and D22 showed clear score advantages and were supported by multiple high-priority user requirements, a full sensitivity analysis of weighting perturbations was not conducted due to the early-stage conceptual nature of the study. Therefore, the exact stability of the ranking should be further verified in future work using a larger expert panel and formal sensitivity analysis. Fifth, the Stateflow simulation validated logical workflow integrity but did not incorporate dynamic physical

parameters such as force feedback, energy consumption, terrain instability, sensor interference, communication delay, dust disturbance, or secondary collapse risk. Therefore, the current validation should be understood as process-level and conceptual-level evidence rather than proof of field readiness. Under real-world disaster conditions, the proposed methodology may still provide a requirement-tracing and decision-support framework, but the final design performance must be further verified through physical prototype testing and controlled field trials.

9.4. Future Directions

Building on the current findings, several directions for future research are identified. Physical prototype development and field testing in simulated or actual earthquake environments would provide empirical validation of the design specifications derived from the proposed framework. Expanding the participant pool to include female rescue personnel and international teams would enhance the generalisability of the experiential data. The integration of digital twin technology with the UXM–Kano–QFD framework could enable real-time requirement updating and adaptive design optimisation. Additionally, extending the methodology to other types of emergency equipment—such as fire rescue apparatus, flood response systems, and hazardous material handling devices—would test the transferability of the proposed approach across different high-risk operational domains.

10. Conclusions

This paper focuses on earthquake demolition rescue robots, addressing challenges in demolition rescue scenarios, including the difficulty of systematically acquiring experiential information, the challenge of quantifying requirement weights, and the lack of effective validation pathways for design decisions. It constructs a systematic design methodology framework integrating UXMs, Kano, QFD, and FBS, and introduces Stateflow workflow simulation and usability evaluation to validate design solutions. This forms a complete methodology chain covering "experiential insight–requirement modelling–Design Translation–Process Validation."

Methodologically, this study first employs UXMs to reconstruct operational workflows and fluctuating user experiences within extreme rescue scenarios such as earthquake debris clearance. This identifies high-load operational nodes and critical experience touchpoints, forming the basis for constructing the HML′ comprehensive requirement set. Subsequently, the Kano model was applied to classify and quantify the contribution of requirement attributes to satisfaction, clarifying the differential roles of must-be, one-dimensional, and attractive requirements in the rescue robot design process. Further, the QFD method was employed to systematically map user requirements to engineering design specifications, determining design element priorities and resource allocation directions. Building on this, the FBS model facilitated the derivation of conceptual design, ensuring clear, logical connections and traceability throughout the design decision-making process.

During validation, Stateflow simulation was used to model the demolition and rescue workflow. This provided process-level evidence regarding task continuity, module coordination, and functional integration. Concurrently, the System Usability Scale (SUS) was applied to subjectively evaluate the overall usability of the design in simulated operational scenarios, providing preliminary process-level and perceived-usability evidence for the conceptual design.

At the application level, the integrated UXM–Kano–QFD–FBS design methodology constructed herein achieves systematic enhancement and improved interpretability in complex equipment design processes through cross-stage information feedback and multi-dimensional validation mechanisms. This provides a replicable design pathway for earth-

quake demolition rescue robots. Although the current validation is limited to conceptual-level evaluation, the methodology may be extended to other high-risk emergency equipment with further empirical testing.

Future research may build upon this foundation by integrating physical prototype test platforms to further validate the performance of demolition rescue robots in real operational environments, while progressively expanding engineering evaluation dimensions such as energy consumption and reliability.

Author Contributions: Conceptualisation, W.P.; methodology, Y.C.; formal analysis, W.P. and Y.H.; investigation, Y.X. and Y.H.; data curation, Y.X.; writing—original draft preparation, Y.X.; writing—review and editing, W.P., Y.X., H.W. and X.L.; visualisation, Y.T. and X.L.; supervision, Y.C.; project administration, H.W. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Hubei University of Technology (grant number 2025XZ14).

Institutional Review Board Statement: This study was approved by the Research Ethics and Technology Security Committee of the School of Industrial Design, Hubei University of Technology (approval date: 1 January 2025; validity period: 1 January 2025 to 31 December 2026). The study was conducted in accordance with the principles of the Declaration of Helsinki.

Informed Consent Statement: Written informed consent was obtained from all participants involved in the UXM experiential study, the Kano questionnaire survey, and the SUS evaluation. Participants were informed of the study objectives, procedures, and their right to withdraw at any time without consequence.

Data Availability Statement: The data presented in this study are available on request from the corresponding author. The data are not publicly available due to participant privacy considerations.

Acknowledgments: During the preparation of this manuscript, the authors used the DeepSeek-V3.2 artificial intelligence tool for translation and language editing. The authors have reviewed and edited the content of this publication and take full responsibility for its accuracy and integrity.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Wang, Y.; Zhang, Y.; Zhang, J.; Li, M.; Ma, M.; Li, J. Multidimensional evaluation of seismic emergency capabilities in Chinese cities: The case of Changchun. *Sci. Rep.* **2024**, *14*, 30898. [CrossRef] [PubMed]
2. Lu, M.; Zhao, X.; Wang, J.; Liu, K.; Li, J.; Xu, J.; Yan, J.; Zhang, X. Analysis of the disaster characteristics of the M6.2 earthquake occurring in Jishishan County, Gansu Province, China, in 2023. *Prog. Earthq. Sci.* **2025**, *55*, 373–382. [CrossRef]
3. Dong, X.; Wang, X. Development of rescue robots and their applications in disaster rescue. *J. Disaster Prev. Mitig. Eng.* **2007**, *27*, 112–117. [CrossRef]
4. Delmerico, J.; Mintchev, S.; Giusti, A.; Gromov, B.; Melo, K.; Horvat, T.; Cadena, C.; Hutter, M.; Ijspeert, A.; Floreano, D.; et al. The current state and future outlook of rescue robotics. *J. Field Robot.* **2019**, *36*, 1171–1191. [CrossRef]
5. Peña Queralta, J.; Taipalmaa, J.; Can Pullinen, B.; Sarker, V.K.; Nguyen Gia, T.; Tenhunen, H.; Gabbouj, M.; Raitoharju, J.; Westerlund, T. Collaborative multi-robot search and rescue: Planning, coordination, perception, and active vision. *IEEE Access* **2020**, *8*, 191617–191643. [CrossRef]
6. Wang, Z. Bionic Design of Earthquake Rescue Excavator. Master's Thesis, Jingdezhen Ceramic University, Jingdezhen, China, 2021. [CrossRef]
7. Zhao, Z. Conceptual Design of Earthquake Rescue Robot Based on Demand and Functional Evolution. Master's Thesis, China University of Mining and Technology, Beijing, China, 2023.
8. Li, X.; Liang, J.; Li, H. Earthquake rescue robot design based on AHP/QFD and TRIZ. *J. Mech. Des.* **2021**, *38*, 121–128. [CrossRef]
9. Li, J.; Kim, K. Kano-QFD-based analysis of the influence of user experience on the design of handicraft intangible cultural heritage apps. *Herit. Sci.* **2023**, *11*, 55. [CrossRef]
10. Kim, Y.S. Customer Experience Design for Smart Product-Service Systems Based on the Iterations of Experience–Evaluate–Engage Using Customer Experience Data. *Sustainability* **2022**, *15*, 686. [CrossRef]

11. Li, L.; Zhao, Z. Comprehensive Performance Evaluation of Earthquake Search and Rescue Robots Based on Improved FAHP and Radar Chart. *Appl. Sci.* **2024**, *14*, 3099. [CrossRef]
12. Gao, J.; Li, Y.; Jin, J.; Jia, Z.; Wei, C. Design, Analysis and Control of Tracked Mobile Robot with Passive Suspension on Rugged Terrain. *Actuators* **2025**, *14*, 389. [CrossRef]
13. Burke, J.L.; Murphy, R.R.; Coovert, M.D.; Riddle, D.L. Moonlight in Miami: Field study of human-robot interaction in the context of an urban search and rescue disaster response training exercise. *Hum.-Comput. Interact.* **2004**, *19*, 85–116. [CrossRef]
14. Asplund, F.; Ulfvengren, P. Engineer-Centred Design Factors and Methodological Approach for Maritime Autonomy Emergency Response Systems. *Safety* **2022**, *8*, 54. [CrossRef]
15. Kalina, D.; O'Neill, R.; Pevere, E.; Fernandez Rojas, R. Operational Fire Management System (OFMS): A Sensor-Integrated Framework for Enhanced Fireground Situational Awareness. *J. Sens. Actuator Netw.* **2025**, *14*, 114. [CrossRef]
16. Gero, J.S.; Kannengiesser, U. The situated function–behaviour–structure framework. *Des. Stud.* **2004**, *25*, 373–391. [CrossRef]
17. Niu, X.; Ye, J.; Chen, L. A Digital Twin-Enhanced KJ-Kano Framework for User-Centric Conceptual Design of Underwater Rescue Robots. *Appl. Sci.* **2025**, *16*, 135. [CrossRef]
18. Ying, F.; Zhai, B.; Zhao, X. Design of a Multi-Method Integrated Intelligent UAV System for Vertical Greening Maintenance. *Appl. Sci.* **2025**, *15*, 887. [CrossRef]
19. Pantouvakis, A.; Gerou, A. The Theoretical and Practical Evolution of Customer Journey and Its Significance in Services Sustainability. *Sustainability* **2022**, *14*, 9610. [CrossRef]
20. Al Rabaiei, K.; Alnajjar, F.; Ahmad, A. Kano Model Integration with Data Mining to Predict Customer Satisfaction. *Big Data Cogn. Comput.* **2021**, *5*, 66. [CrossRef]
21. Lee, C.-H.; Chen, C.-H.; Lin, C.; Li, F.; Zhao, X. Developing a Quick Response Product Configuration System under Industry 4.0 Based on Customer Requirement Modelling and Optimization Method. *Appl. Sci.* **2019**, *9*, 4. [CrossRef]
22. Shi, H.; Mao, L.-X.; Li, K.; Wang, X.-H.; Liu, H.-C. Engineering Characteristics Prioritization in Quality Function Deployment Using an Improved ORESTE Method with Double Hierarchy Hesitant Linguistic Information. *Sustainability* **2022**, *14*, 9771. [CrossRef]
23. Raimbaud, P.; Lou, R.; Danglade, F.; Figueroa, P.; Hernandez, J.T.; Merienne, F. A Task-Centred Methodology to Evaluate the Design of Virtual Reality User Interactions: A Case Study on Hazard Identification. *Buildings* **2021**, *11*, 277. [CrossRef]
24. Jacoff, A.; Messina, E.; Weiss, B.A.; Tadokoro, S.; Nakagawa, Y. Test arenas and performance metrics for urban search and rescue robots. In Proceedings of the 2003 IEEE/RSJ International Conference on Intelligent Robots and Systems, Las Vegas, NV, USA, 27–31 October 2003; pp. 3396–3403. [CrossRef]
25. DeJonckheere, M.; Vaughn, L.M. Semistructured interviewing in primary care research: A balance of relationship and rigour. *Fam. Med. Community Health* **2019**, *7*, e000057. [CrossRef] [PubMed]
26. Hart, S.G.; Staveland, L.E. Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research. In *Human Mental Workload*; Hancock, P.A., Meshkati, N., Eds.; North-Holland: Amsterdam, The Netherlands, 1988; pp. 139–183. [CrossRef]
27. Surmann, H.; Daun, K.; Schnaubelt, M.; von Stryk, O.; Patchou, M.; Böcker, S.; Wietfeld, C.; Quenzel, J.; Schleich, D.; Behnke, S.; et al. Lessons from robot-assisted disaster response deployments by the German Rescue Robotics Center task force. *J. Field Robot.* **2023**, *41*, 782–797. [CrossRef]
28. Ai, X.; Jiang, Z.; Hu, K.; Chandrasekaran, S.; Wang, Y. Integrating a Cross-Reference List and Customer Journey Map to Improve Industrial Design Teaching and Learning in "Project-Oriented Design Based Learning". *Sustainability* **2020**, *12*, 4672. [CrossRef]
29. Wang, T.-C.; Guo, R.-S.; Chen, C.; Li, C.-K. Multi-Stage Data-Driven Framework for Customer Journey Optimization and Operational Resilience. *Mathematics* **2025**, *13*, 1145. [CrossRef]
30. Yang, C.-M.; Li, S.; Chen, K.-S.; Li, M.; Lo, W. A Methodological Framework for New Product Development in Fuzzy Environments. *Systems* **2024**, *12*, 382. [CrossRef]
31. Lee, A.H.I.; Lai, C.-J.; Kang, H.-Y.; Wang, C.-C. An Integrated Fuzzy Quality Function Deployment Model for Designing Touch Panels. *Mathematics* **2025**, *13*, 2636. [CrossRef]
32. Haber, N.; Fargnoli, M.; Sakao, T. Integrating QFD for product-service systems with the Kano model and fuzzy AHP. *Total Qual. Manag. Bus. Excell.* **2018**, *31*, 929–954. [CrossRef]
33. Li, Y.; Ghazilla, R.A.R.; Abdul-Rashid, S.H. QFD-Based Research on Sustainable User Experience Optimization Design of Smart Home Products for the Elderly: A Case Study of Smart Refrigerators. *Int. J. Environ. Res. Public Health* **2022**, *19*, 13742. [CrossRef]
34. Kim, T.H.; Bae, S.H.; Han, C.H.; Hahn, B. The Design of a Low-Cost Sensing and Control Architecture for a Search and Rescue Assistant Robot. *Machines* **2023**, *11*, 329. [CrossRef]
35. Dindorf, R.; Wos, P. Challenges of Robotic Technology in Sustainable Construction Practice. *Sustainability* **2024**, *16*, 5500. [CrossRef]
36. Dai, Y.; Xiang, C.-F.; Liu, Z.-X.; Li, Z.-L.; Qu, W.-Y.; Zhang, Q.-H. Modular Robotic Design and Reconfiguring Path Planning. *Appl. Sci.* **2022**, *12*, 723. [CrossRef]
37. Papazoglou, K. Stress, Prevention, and Resilience among First Responders. *Int. J. Environ. Res. Public Health* **2023**, *20*, 7174. [CrossRef]

38. Rennpferdt, C.; Zuefle, M.; Bagusat, M.; Bender, D.; Krause, D. Development of Modular Architectures for Product–Service Systems. *Sustainability* **2023**, *15*, 14001. [CrossRef]
39. Macris, A.; Drafta, S.; Martiniuc, S.; Petre, A.E. Comparative Usability Evaluation of Three Digital Smile Design Software Tools Using the System Usability Scale. *Dent. J.* **2025**, *13*, 418. [CrossRef] [PubMed]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.

Reproduced with permission of copyright owner. Further reproduction prohibited without permission.